ISSS608-VAA

Hands-on Exercise 1

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Getting Started

Install and launching R packages

The code chunk below uses p_load() of pacman package to check if tidyverse packages are installed in the computer. If they are, then they will be launched into R.

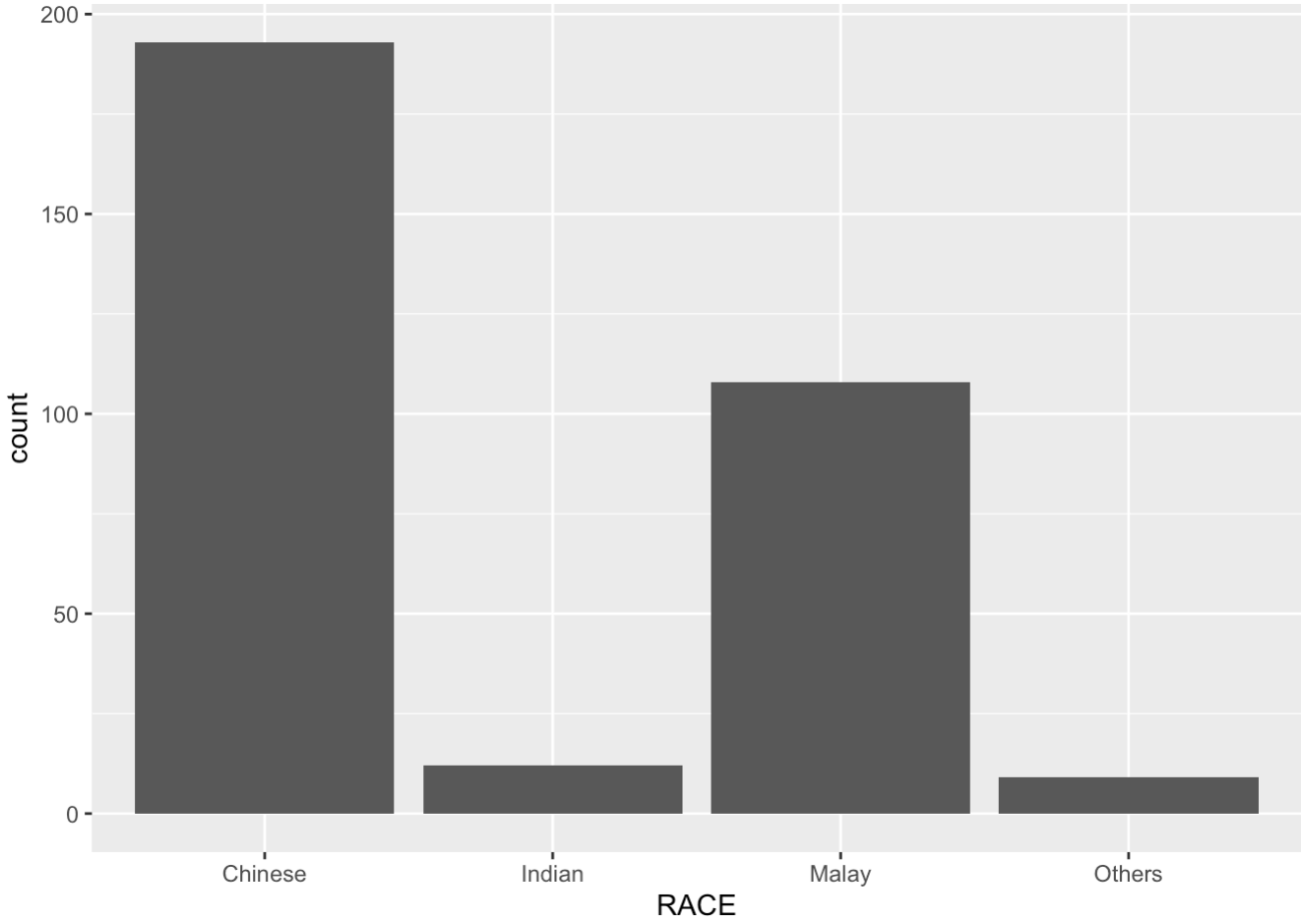
```
pacman::p_load(tidyverse)
```

Importing the data

```
exam_data <- read_csv("Exam_data.csv")
```

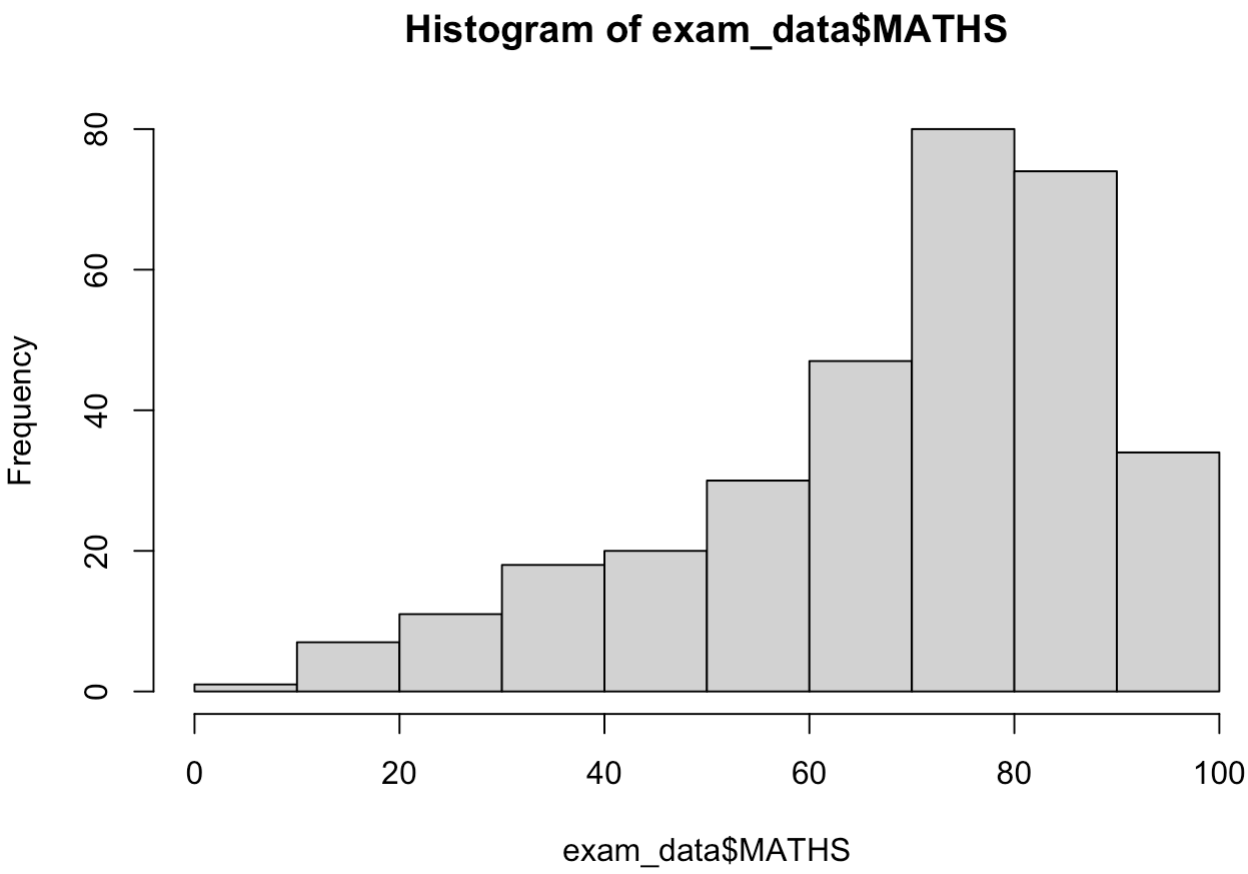
```
Rows: 322 Columns: 7
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (4): ID, CLASS, GENDER, RACE
dbl (3): ENGLISH, MATHS, SCIENCE

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Plotting a simple bar chart



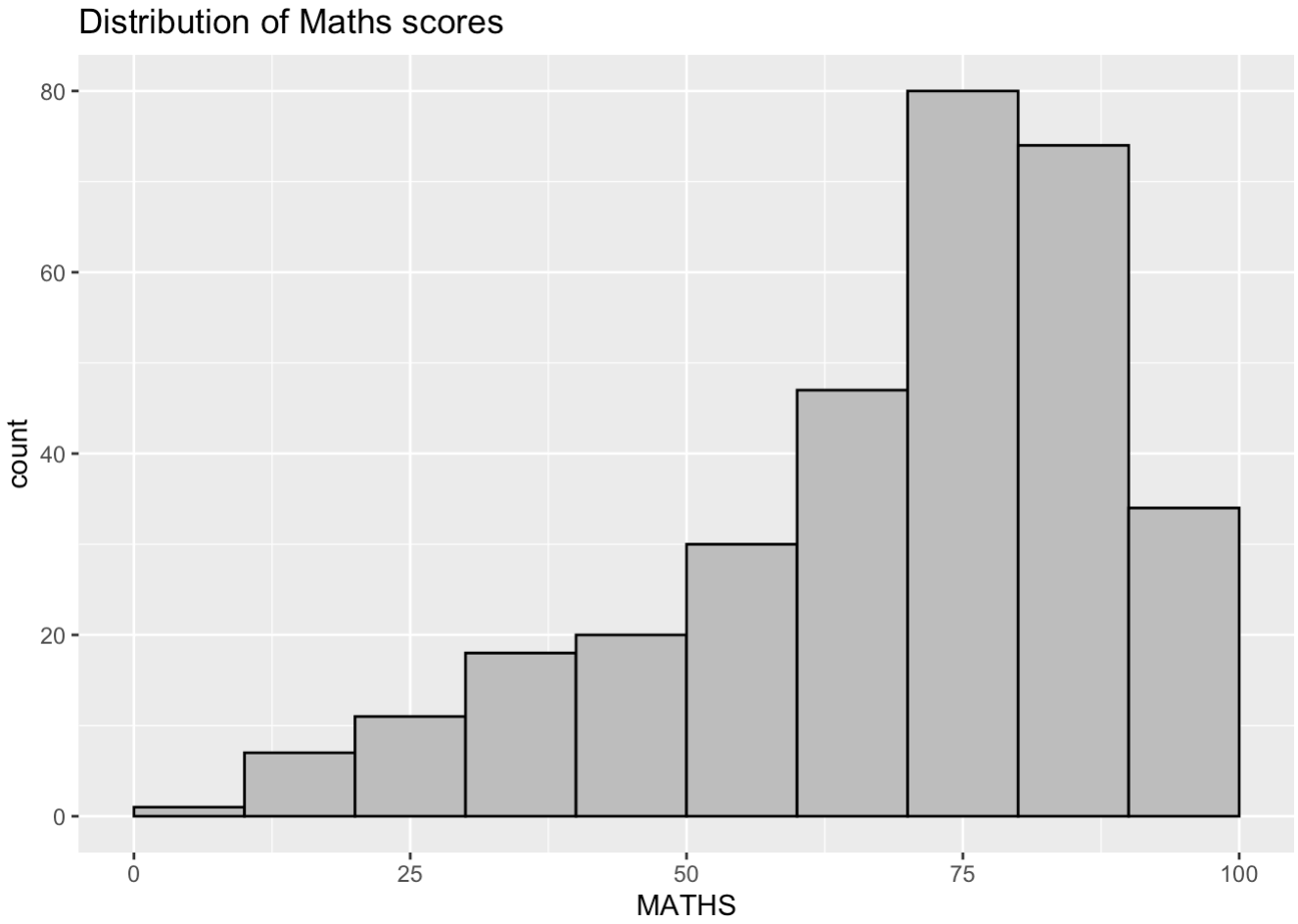
R Graphics histogram:

```
hist(exam_data$MATHS)
```



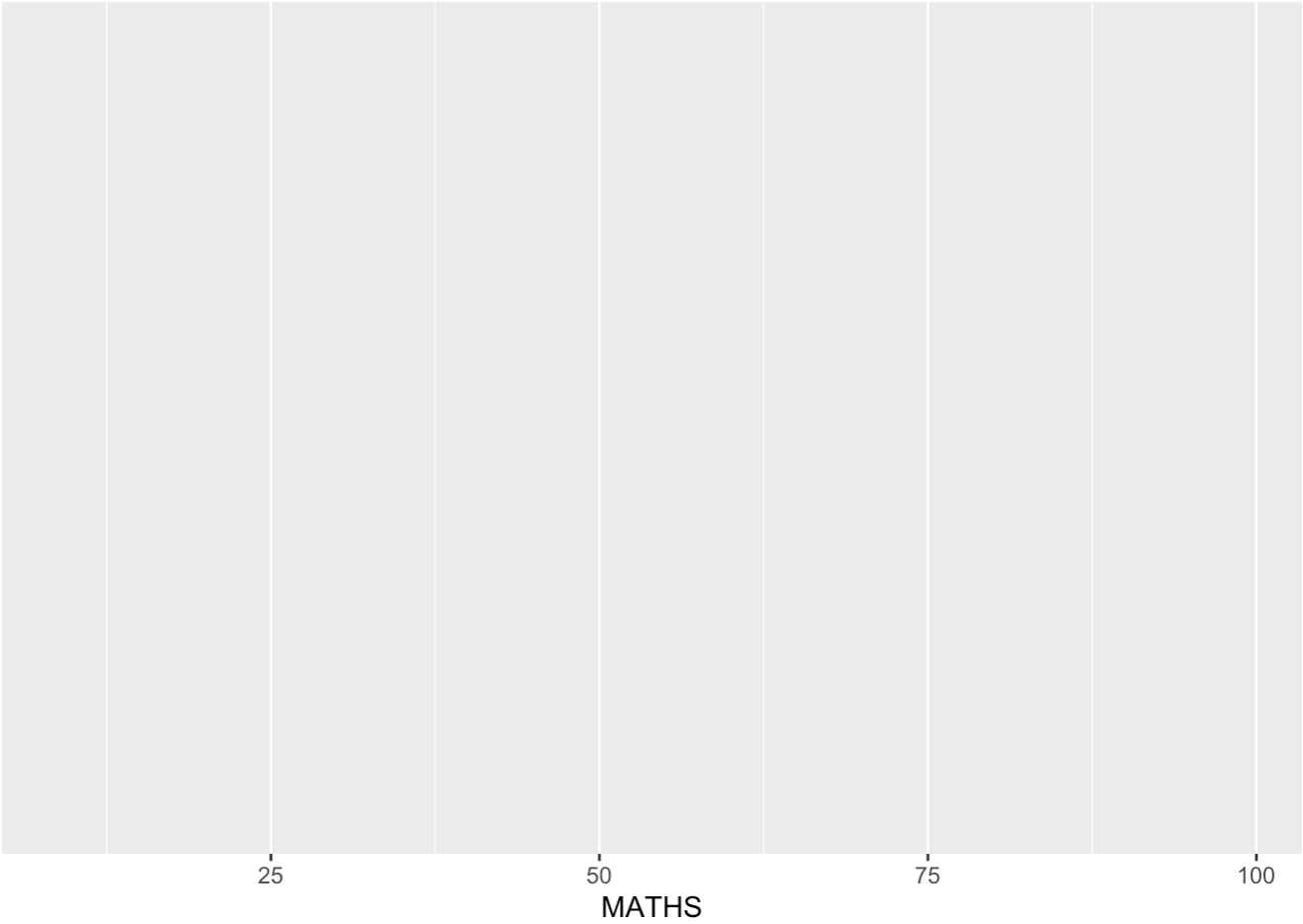
ggplot2 histogram:

```
ggplot(data=exam_data, aes(x = MATHS)) +
  geom_histogram(bins=10,
                 boundary = 100,
                 color="black",
                 fill="grey") +
  ggtitle("Distribution of Maths scores")
```



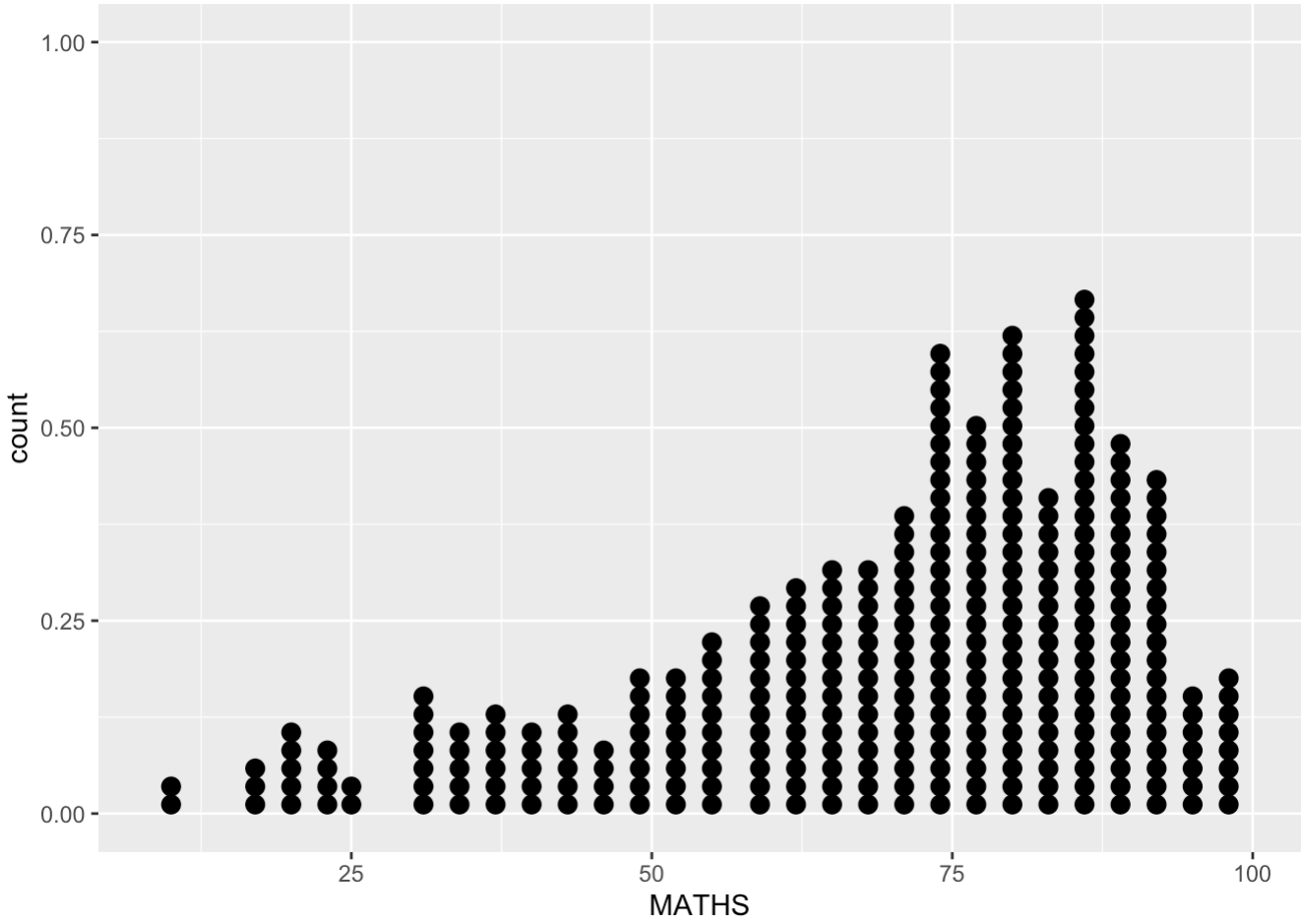

```
ggplot(data=exam_data)
```

```
ggplot(data=exam_data,
       aes(x= MATHS))
```

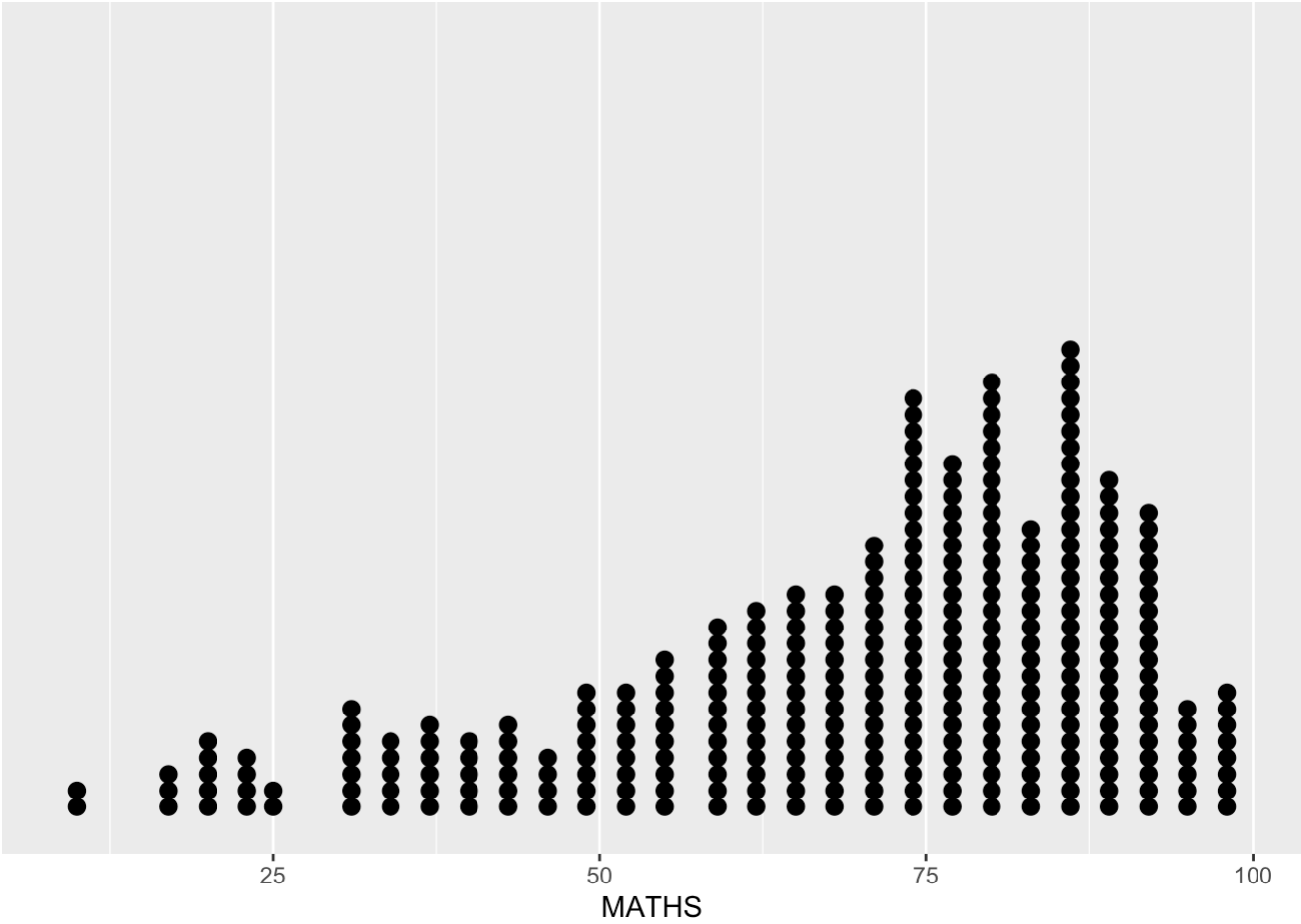



```
ggplot(data=exam_data,
       aes(x = MATHS)) +
  geom_dotplot(dotsize = 0.5)
```

```
Bin width defaults to 1/30 of the range of the data. Pick better value with
`binwidth`.
```

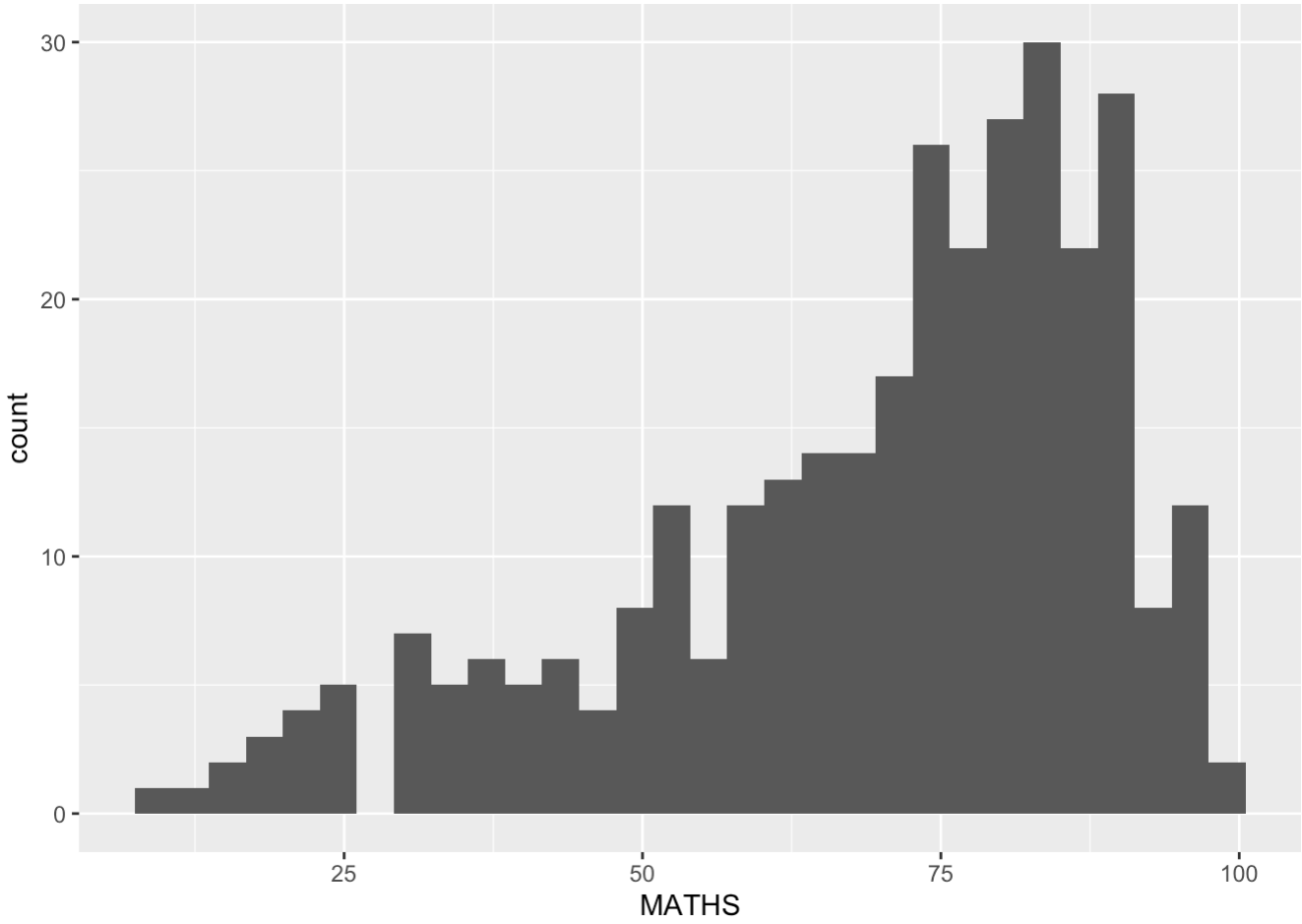



```
ggplot(data=exam_data,
       aes(x = MATHS)) +
  geom_dotplot(binwidth=2.5,
               dotsize = 0.5) +
  scale_y_continuous(NULL,
                     breaks = NULL)
```

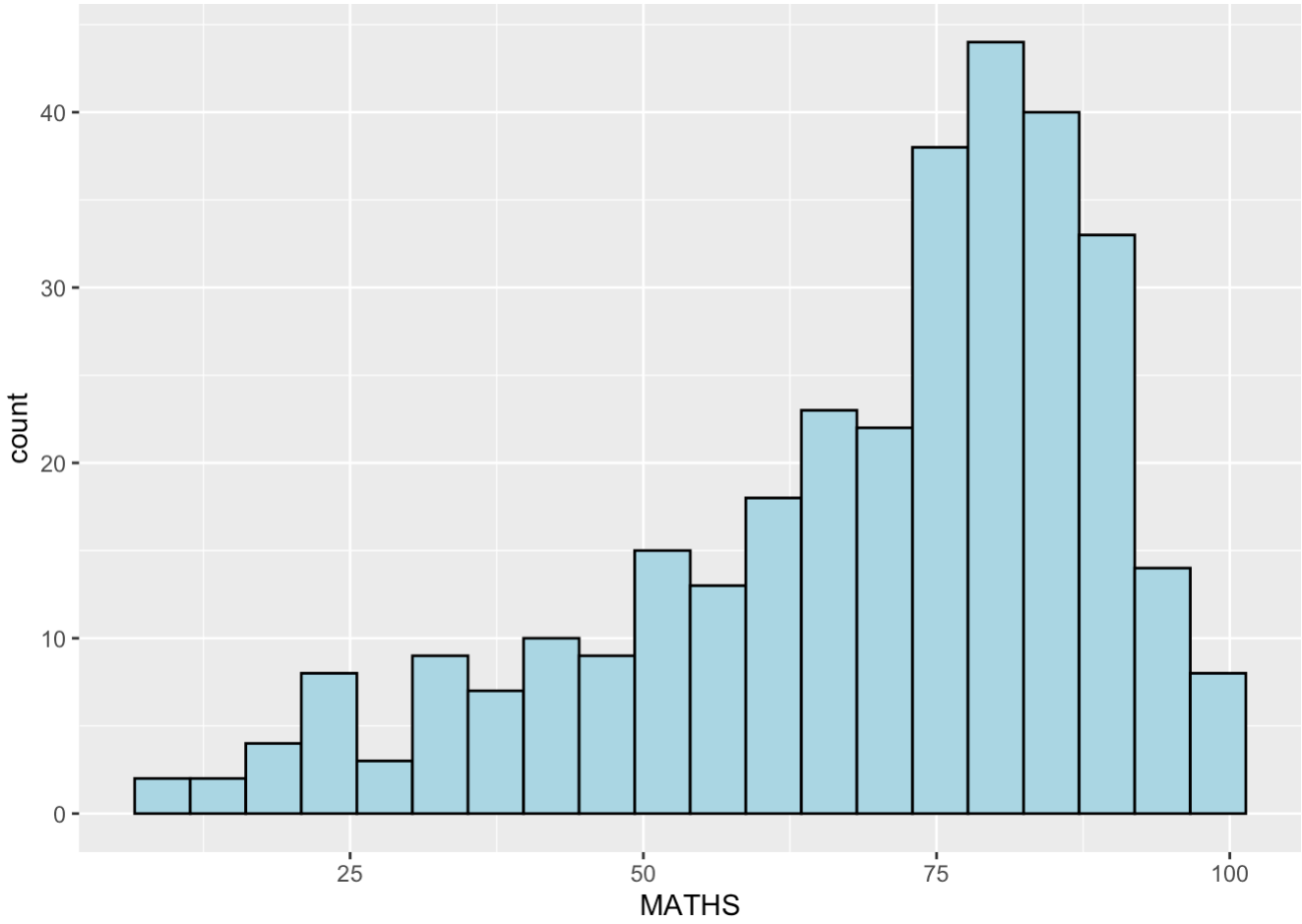



```
ggplot(data=exam_data,
       aes(x = MATHS)) +
  geom_histogram()
```

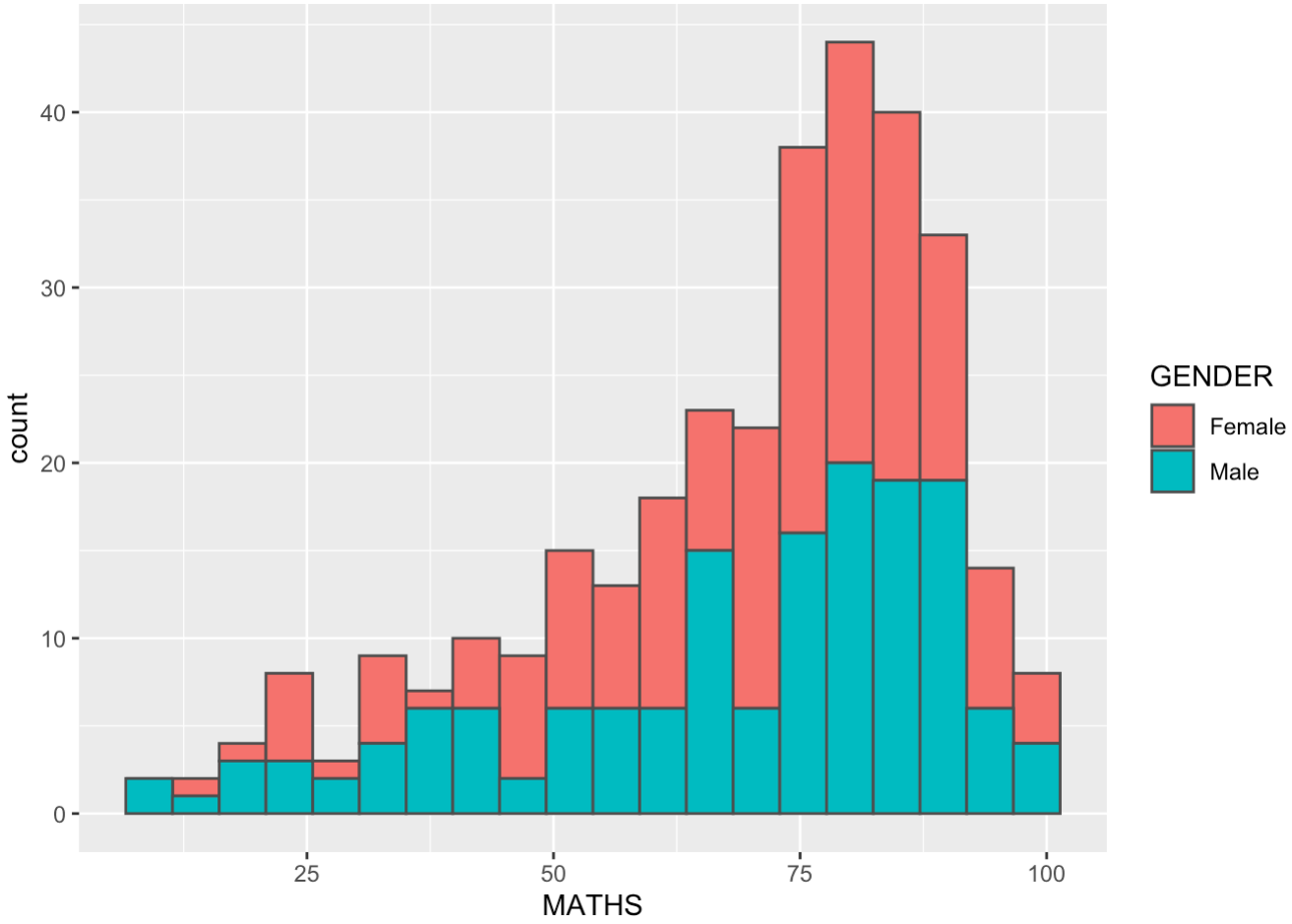
```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ggplot(data=exam_data,
       aes(x= MATHS)) +
  geom_histogram(bins=20,
                 color="black",
                 fill="light blue")
```

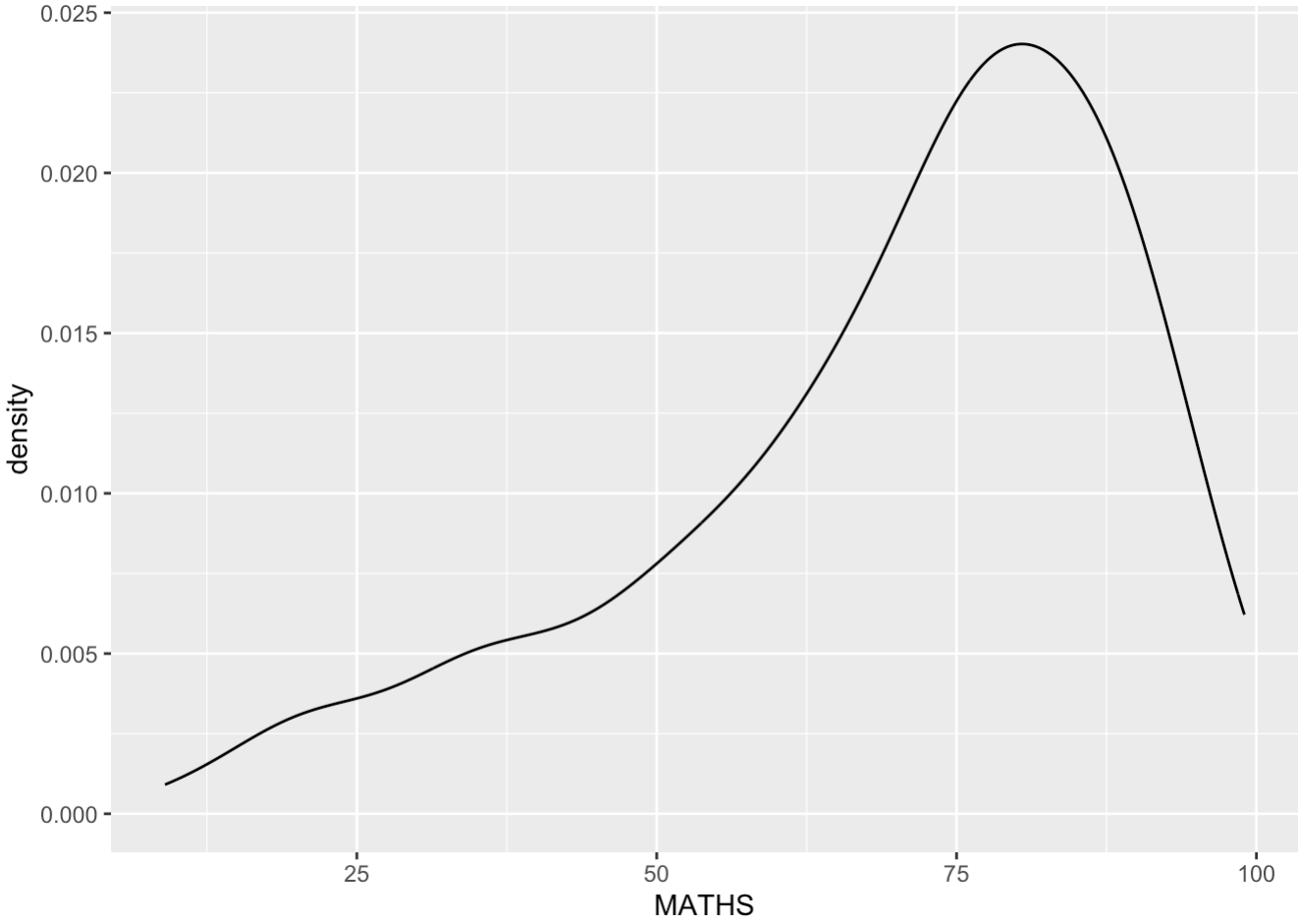



```
ggplot(data=exam_data,
       aes(x= MATHS,
           fill = GENDER)) +
  geom_histogram(bins=20,
                 color="grey30")
```



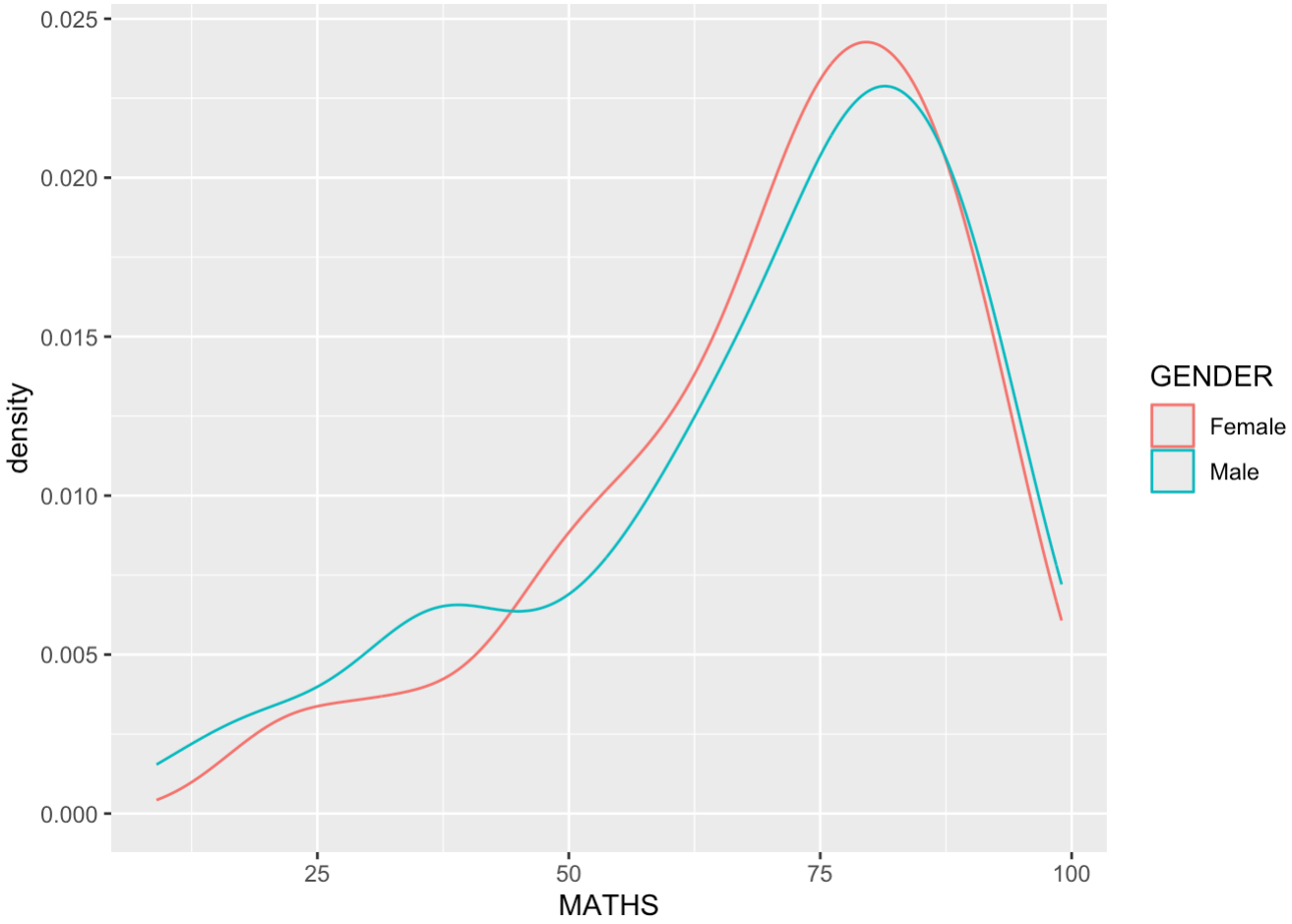
Basic density plot

```
ggplot(data=exam_data,
       aes(x = MATHS)) +
  geom_density()
```



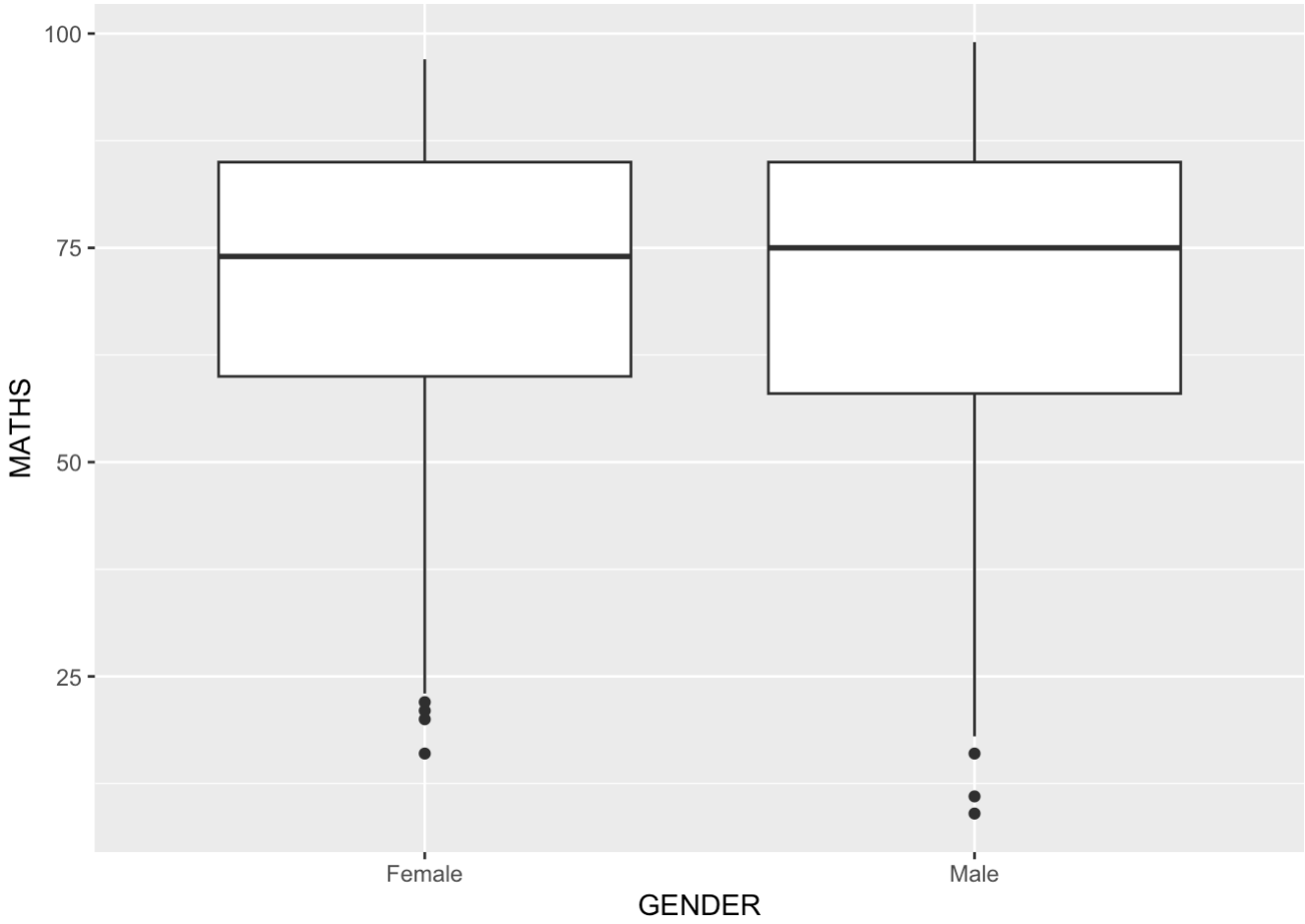
Density by gender

```
ggplot(data=exam_data,
       aes(x = MATHS,
           colour = GENDER)) +
  geom_density()
```



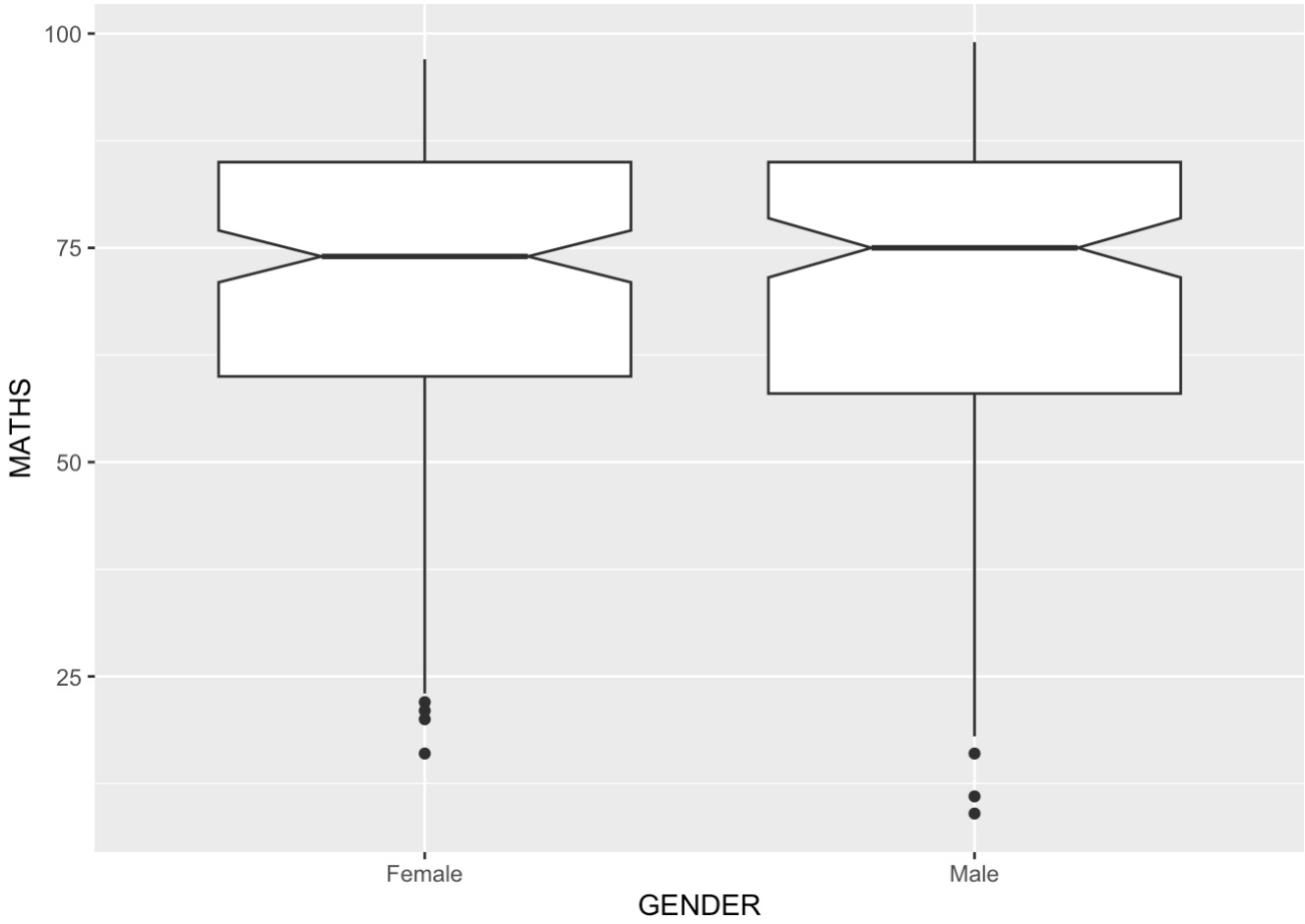
Basic boxplot

```
ggplot(data=exam_data,
       aes(y = MATHS,
           x= GENDER)) +
  geom_boxplot()
```

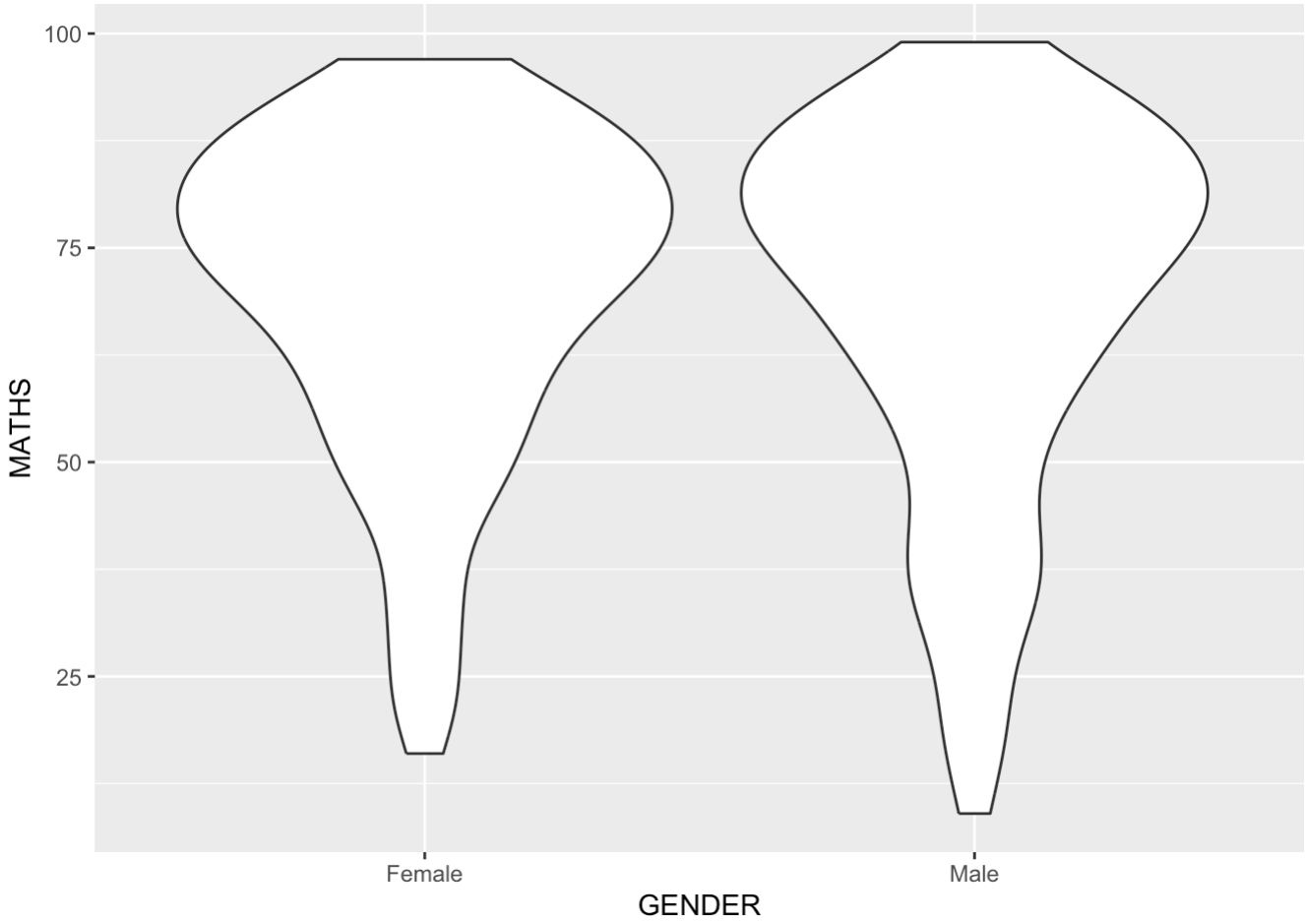


Notched boxplot

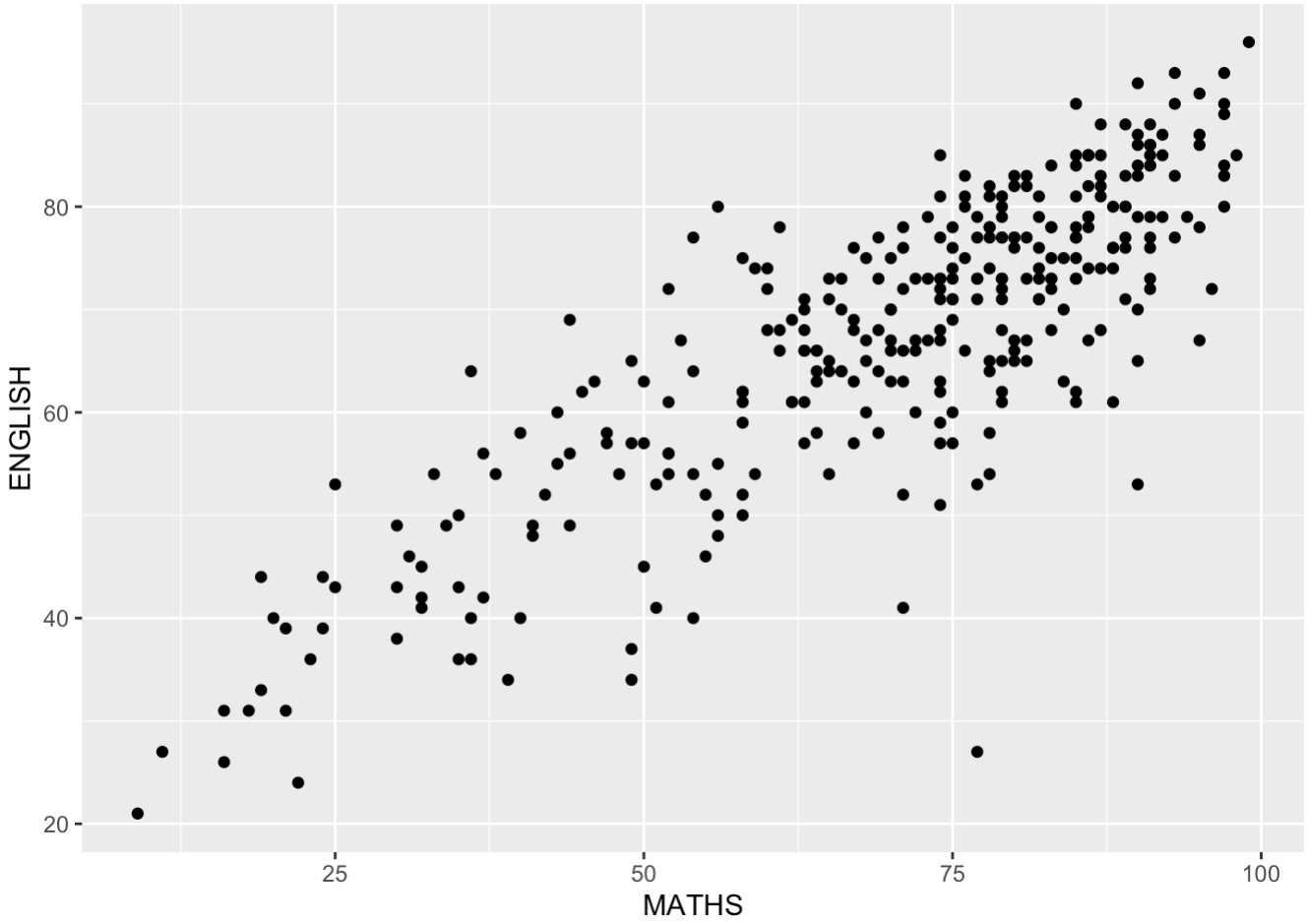
```
ggplot(data=exam_data,
       aes(y = MATHS,
           x= GENDER)) +
  geom_boxplot(notch=TRUE)
```



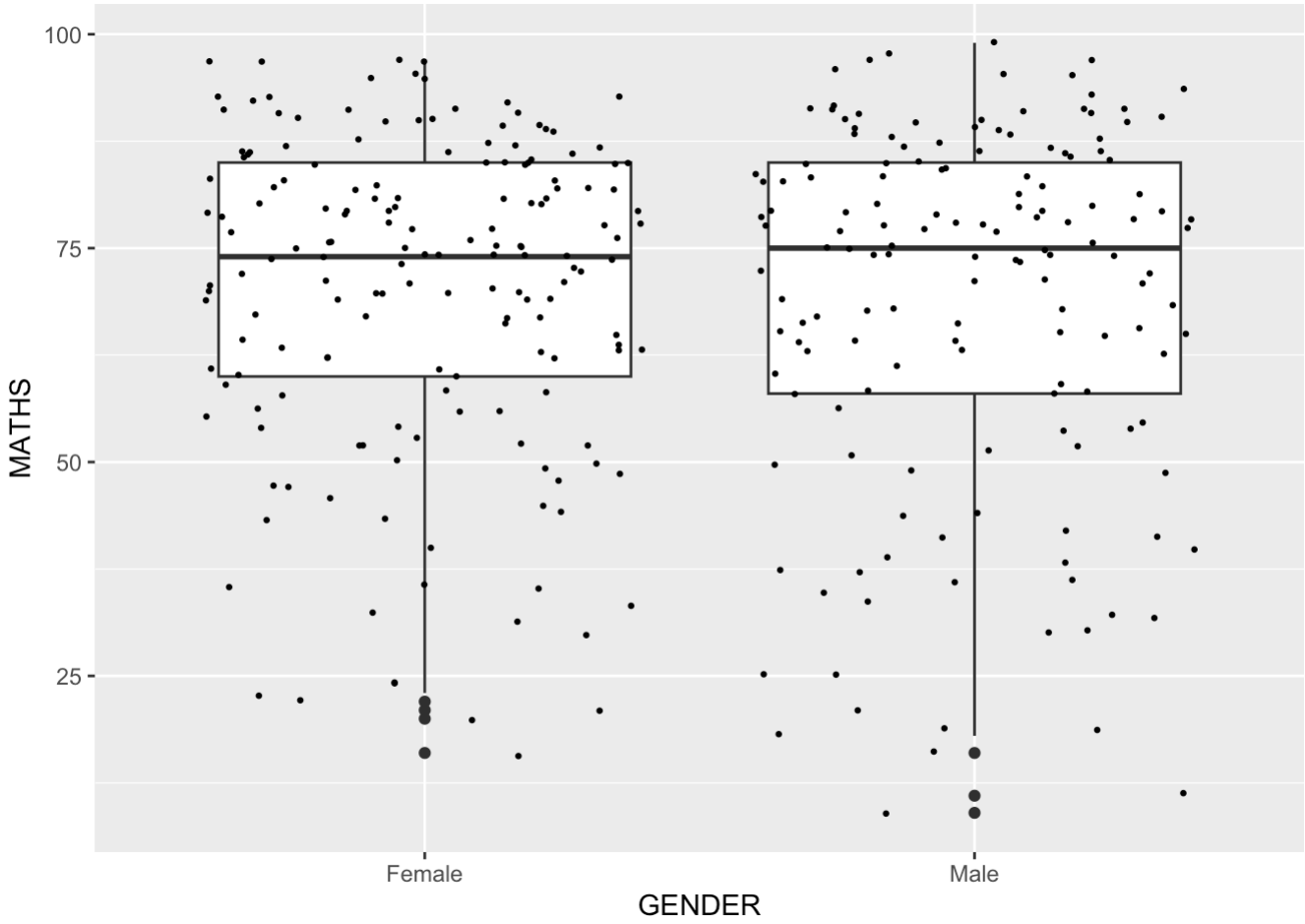

```
ggplot(data=exam_data,
       aes(y = MATHS,
           x= GENDER)) +
  geom_violin()
```



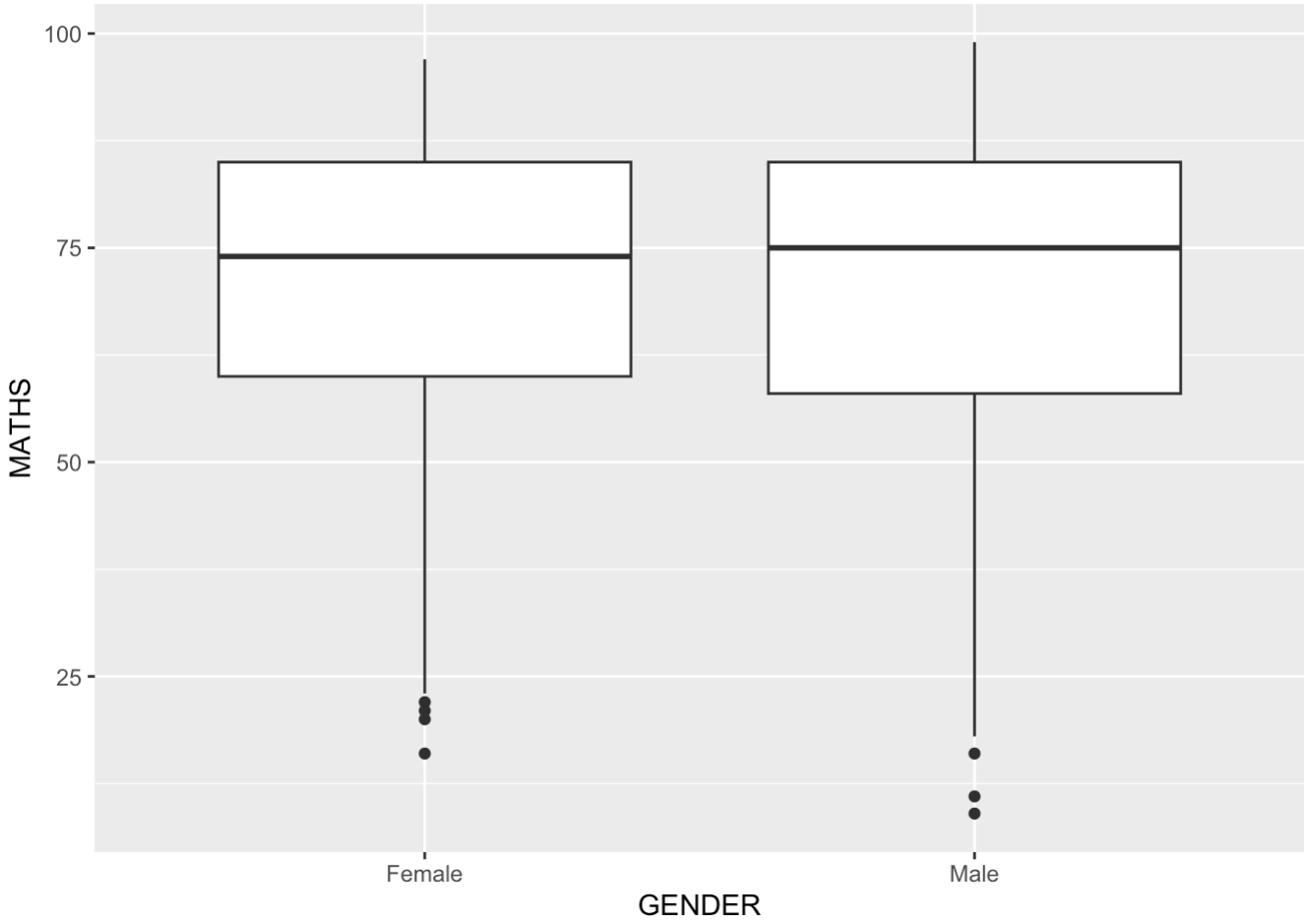

```
ggplot(data=exam_data,
       aes(x= MATHS,
           y=ENGLISH)) +
  geom_point()
```



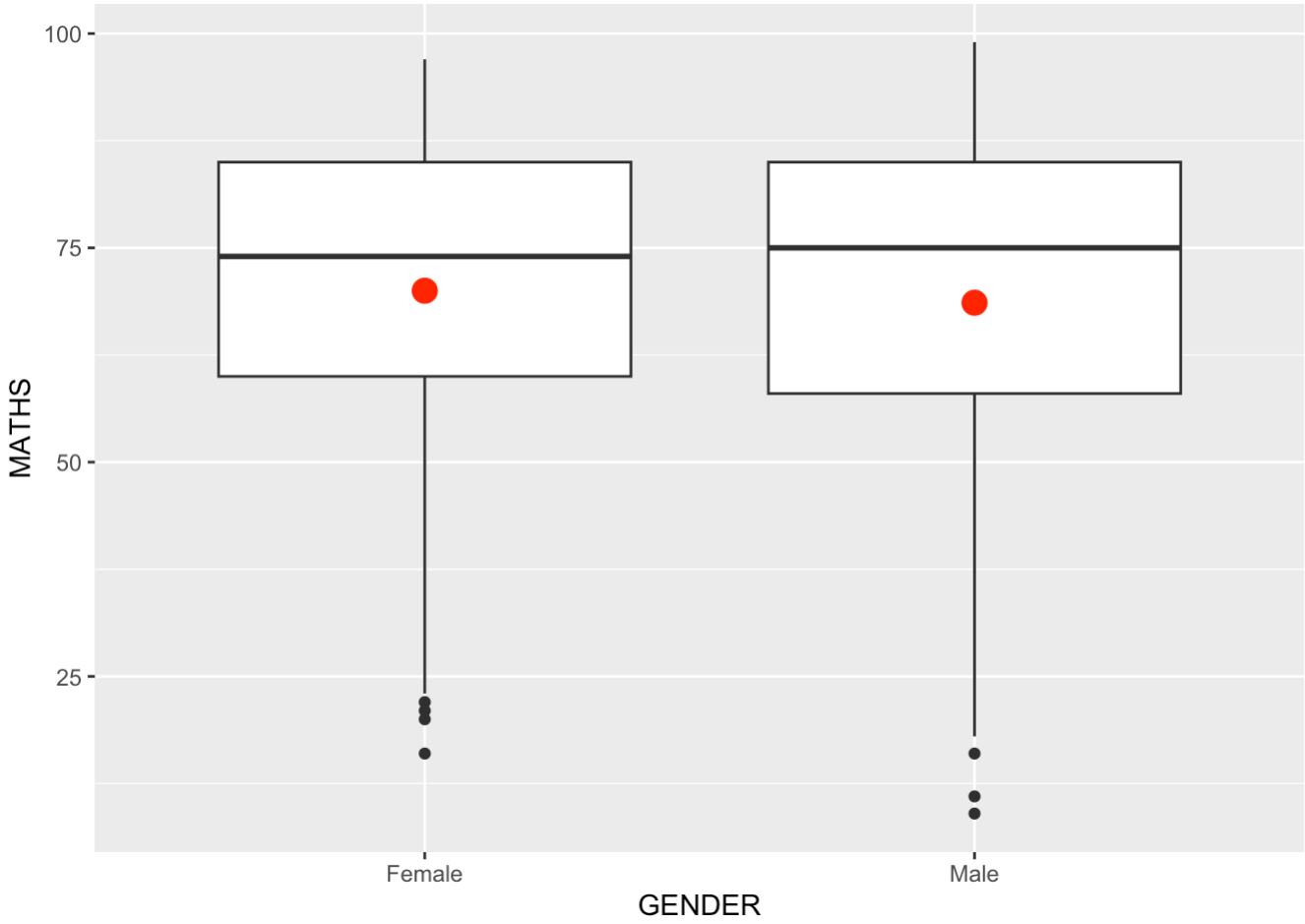

```
ggplot(data=exam_data,
       aes(y = MATHS,
           x= GENDER)) +
  geom_boxplot() +
  geom_point(position="jitter",
             size = 0.5)
```




```
ggplot(data=exam_data,
       aes(y = MATHS, x= GENDER)) +
  geom_boxplot()
```

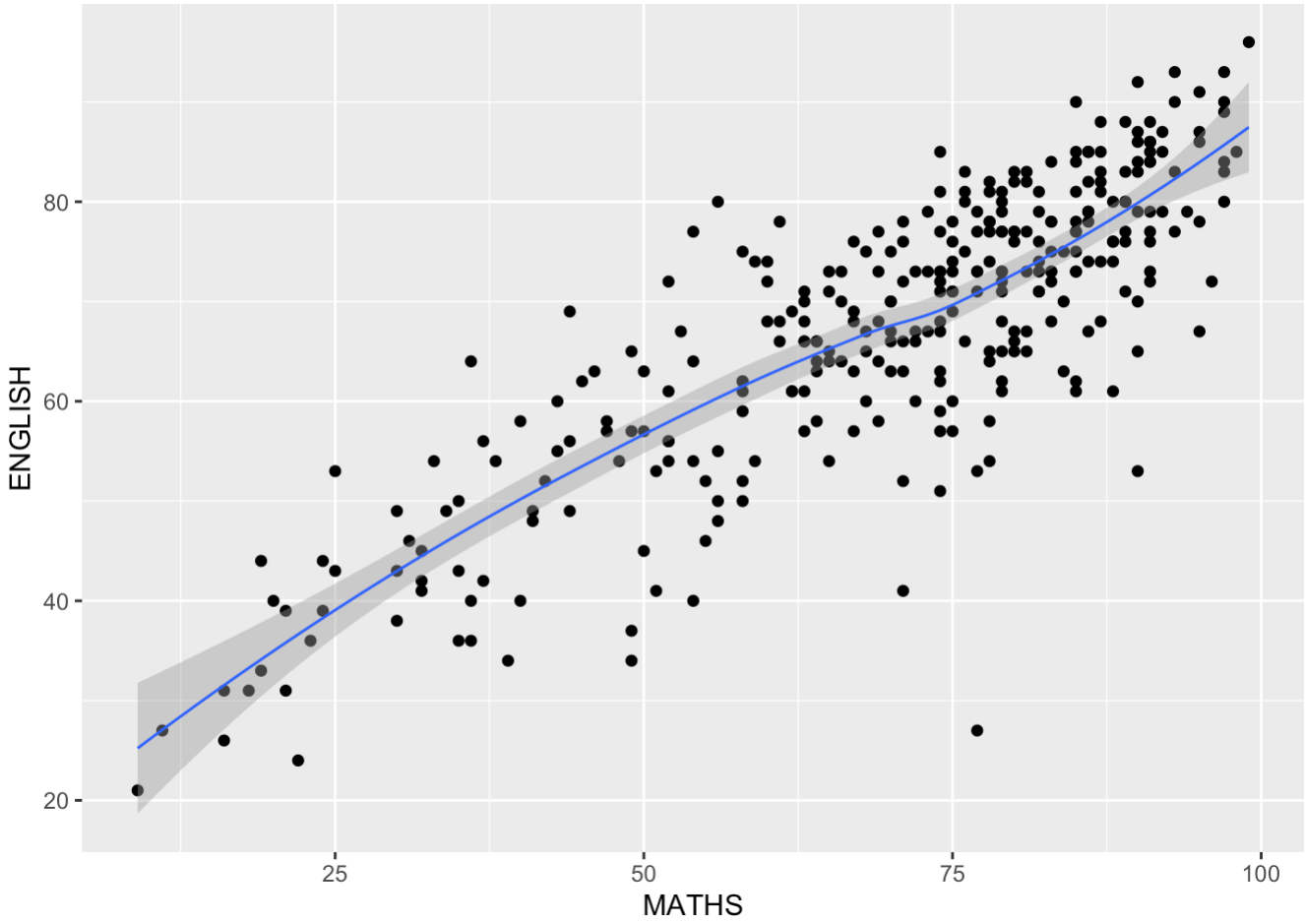



```
ggplot(data=exam_data,
       aes(y = MATHS, x= GENDER)) +
  geom_boxplot() +
  stat_summary(geom = "point",
               fun = "mean",
               colour ="red",
               size=4)
```



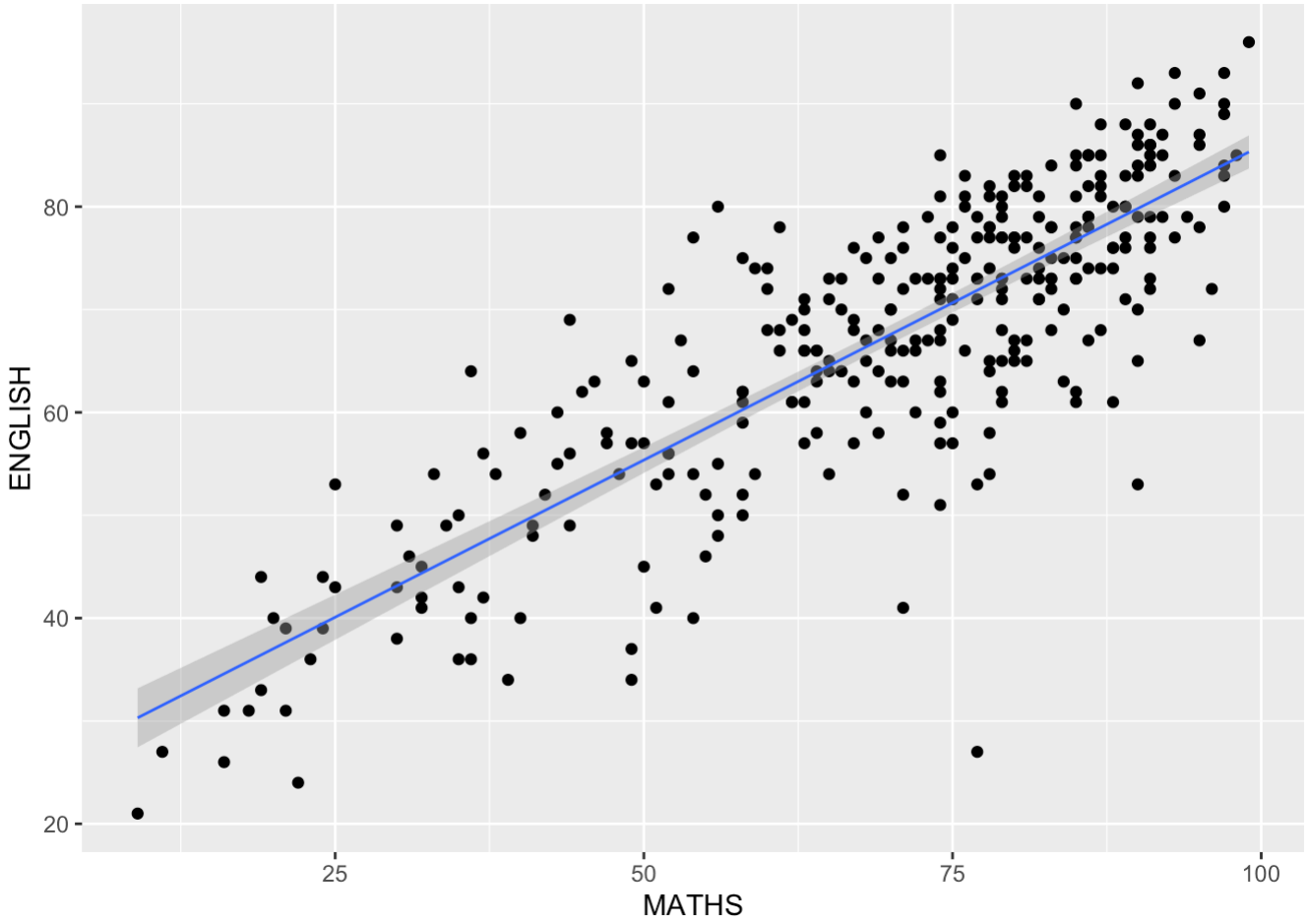

```
ggplot(data=exam_data,
       aes(x= MATHS, y=ENGLISH)) +
  geom_point() +
  geom_smooth(size=0.5)
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
ℹ Please use `linewidth` instead.

`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



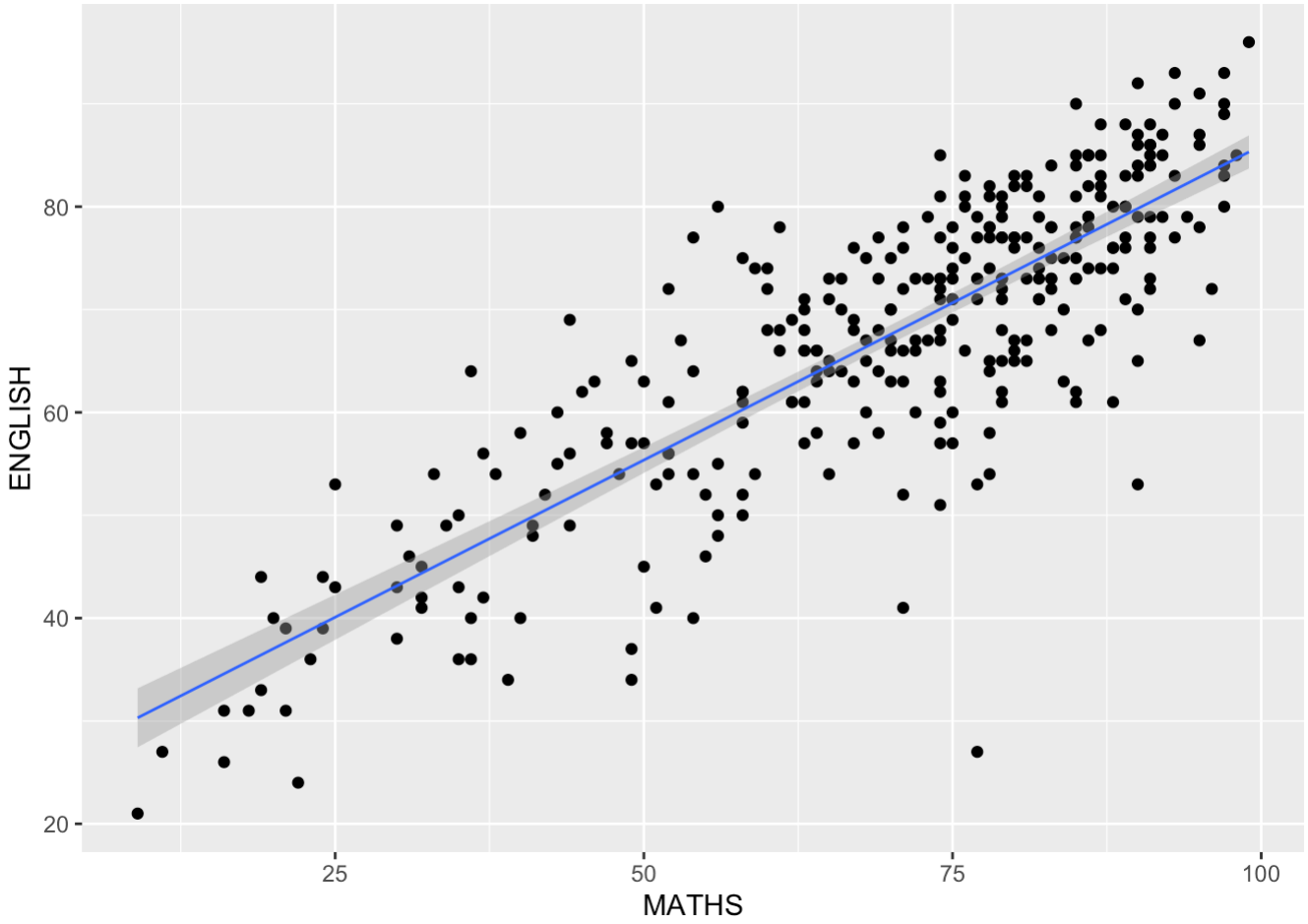

```
ggplot(data=exam_data,
       aes(x= MATHS,
           y=ENGLISH)) +
  geom_point() +
  geom_smooth(method=lm,
              linewidth=0.5)
```

```
`geom_smooth()` using formula = 'y ~ x'
```



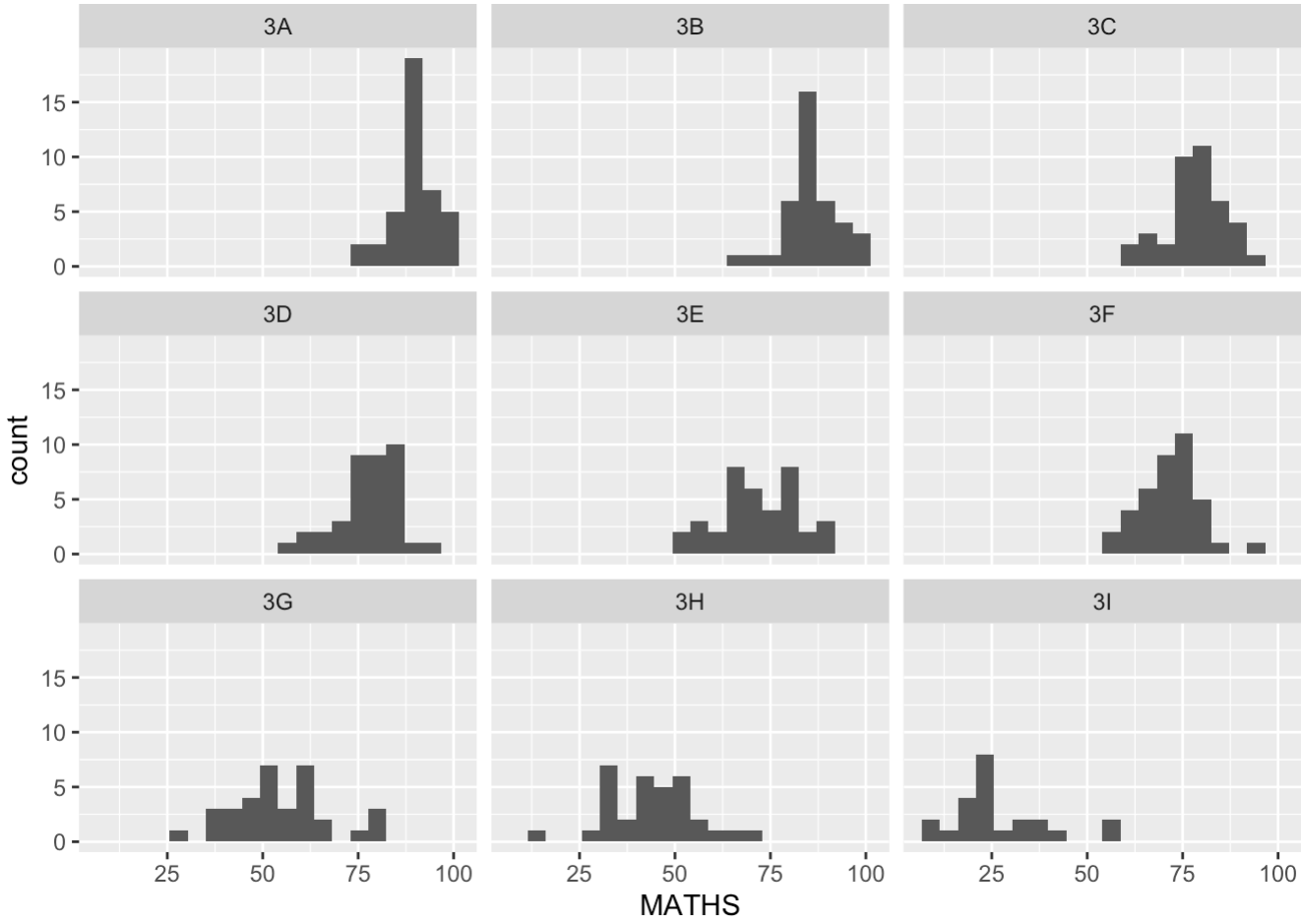

```
ggplot(data=exam_data,
       aes(x= MATHS,
           y=ENGLISH)) +
  geom_point() +
  geom_smooth(method=lm,
              linewidth=0.5)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

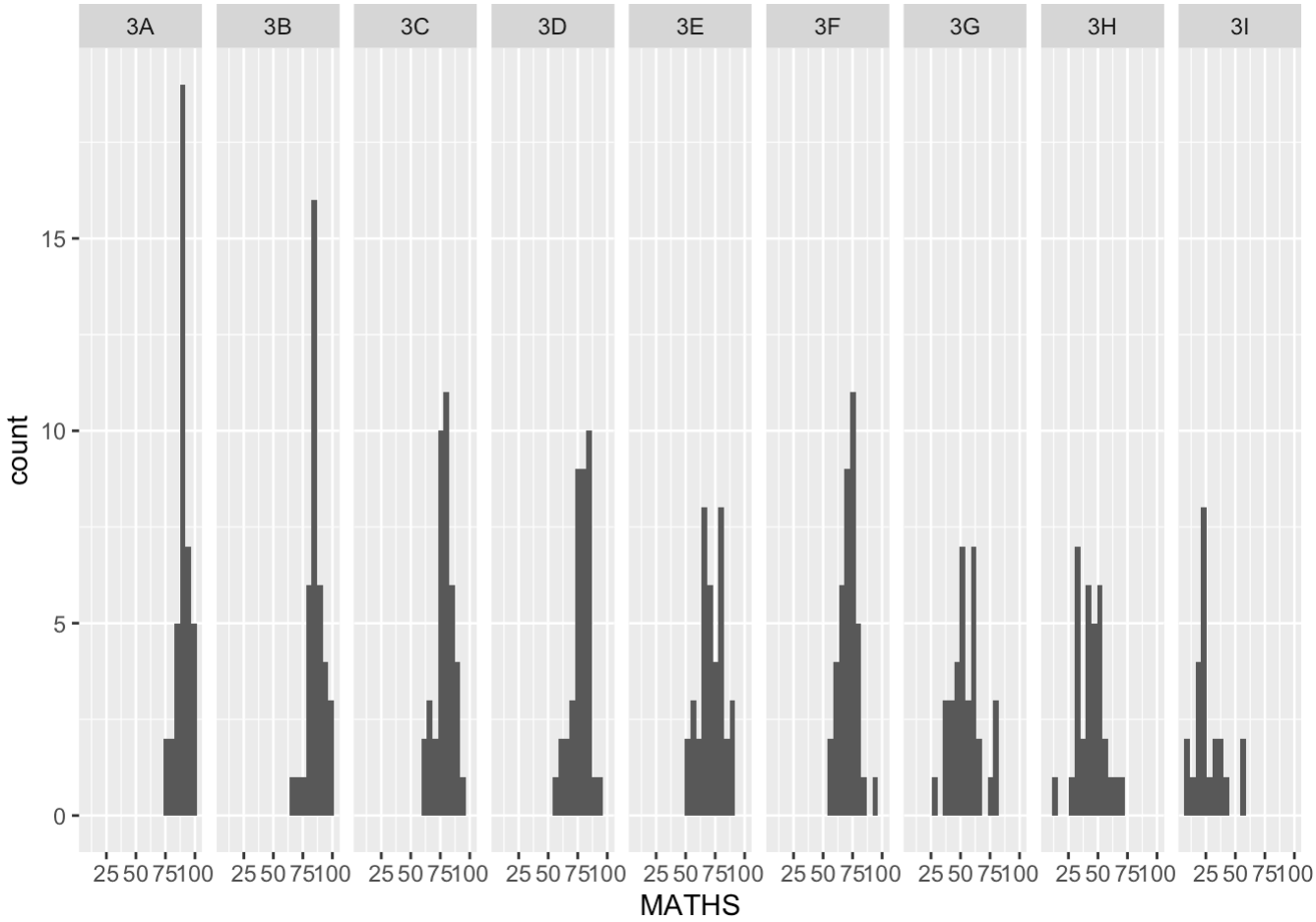


Facets

```
ggplot(data=exam_data,
       aes(x= MATHS)) +
  geom_histogram(bins=20) +
    facet_wrap(~ CLASS)
```

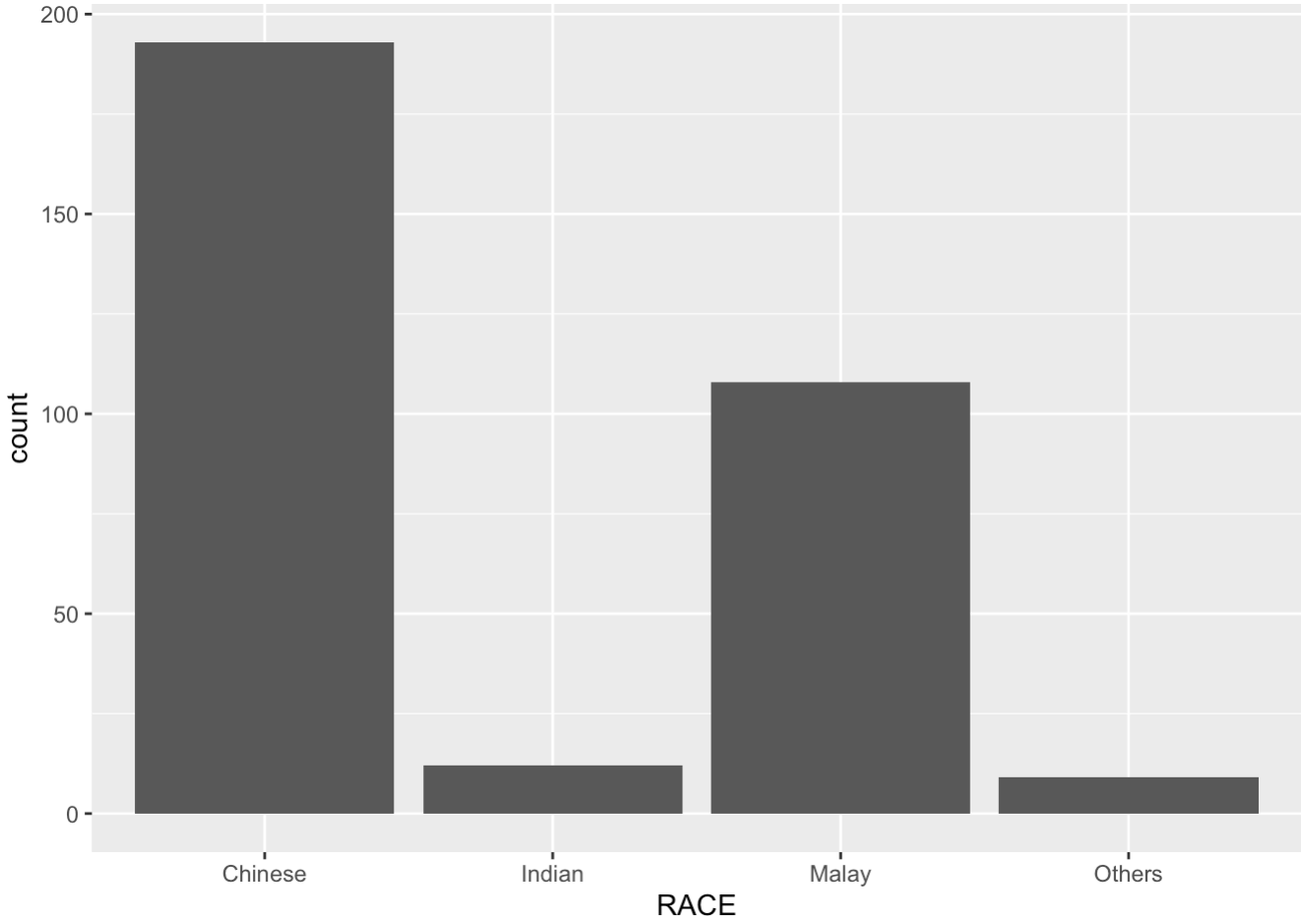



```
ggplot(data=exam_data,
       aes(x= MATHS)) +
  geom_histogram(bins=20) +
    facet_grid(~ CLASS)
```



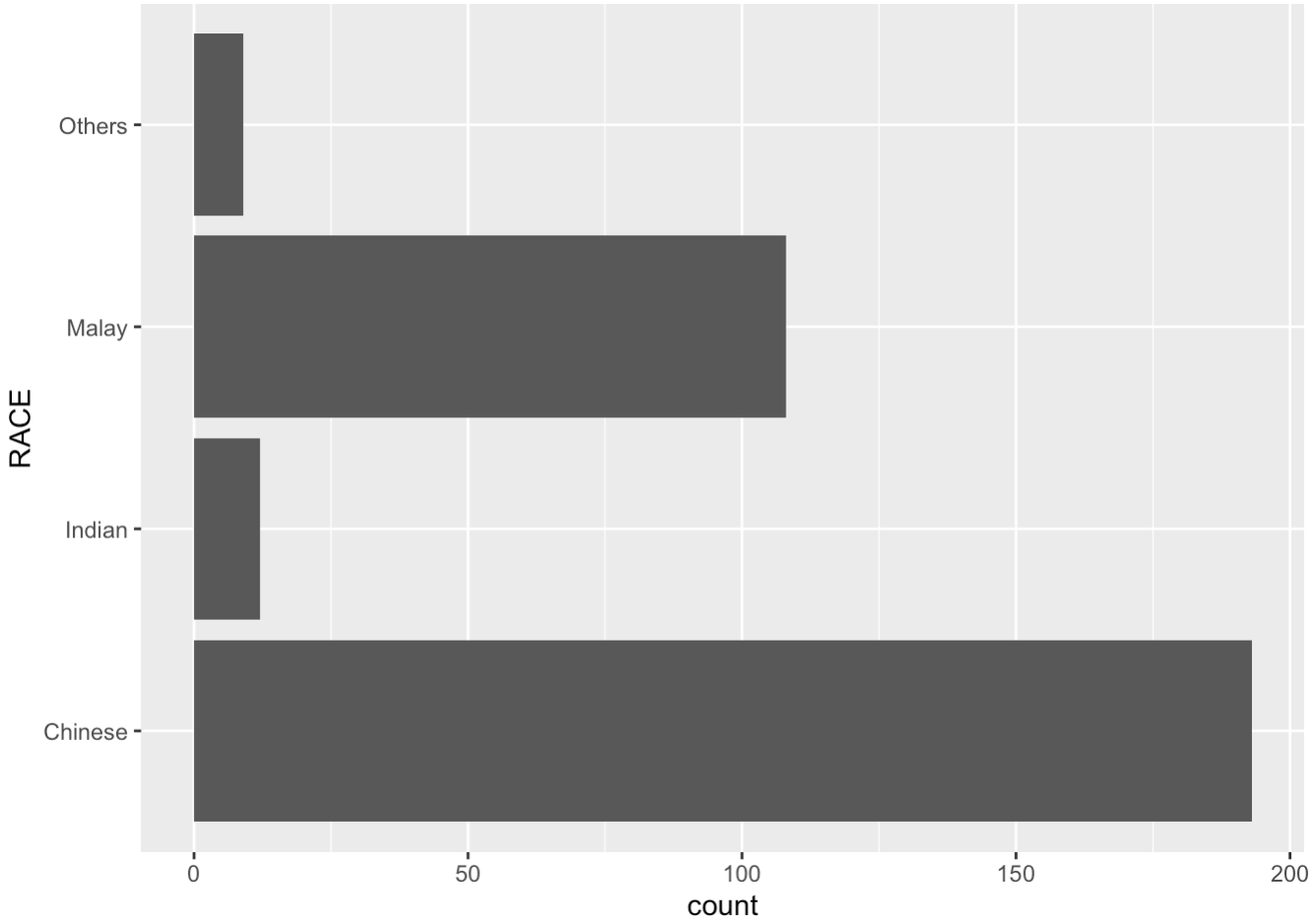
Basic bar chart (vertical)

```
ggplot(data=exam_data,
       aes(x=RACE)) +
  geom_bar()
```



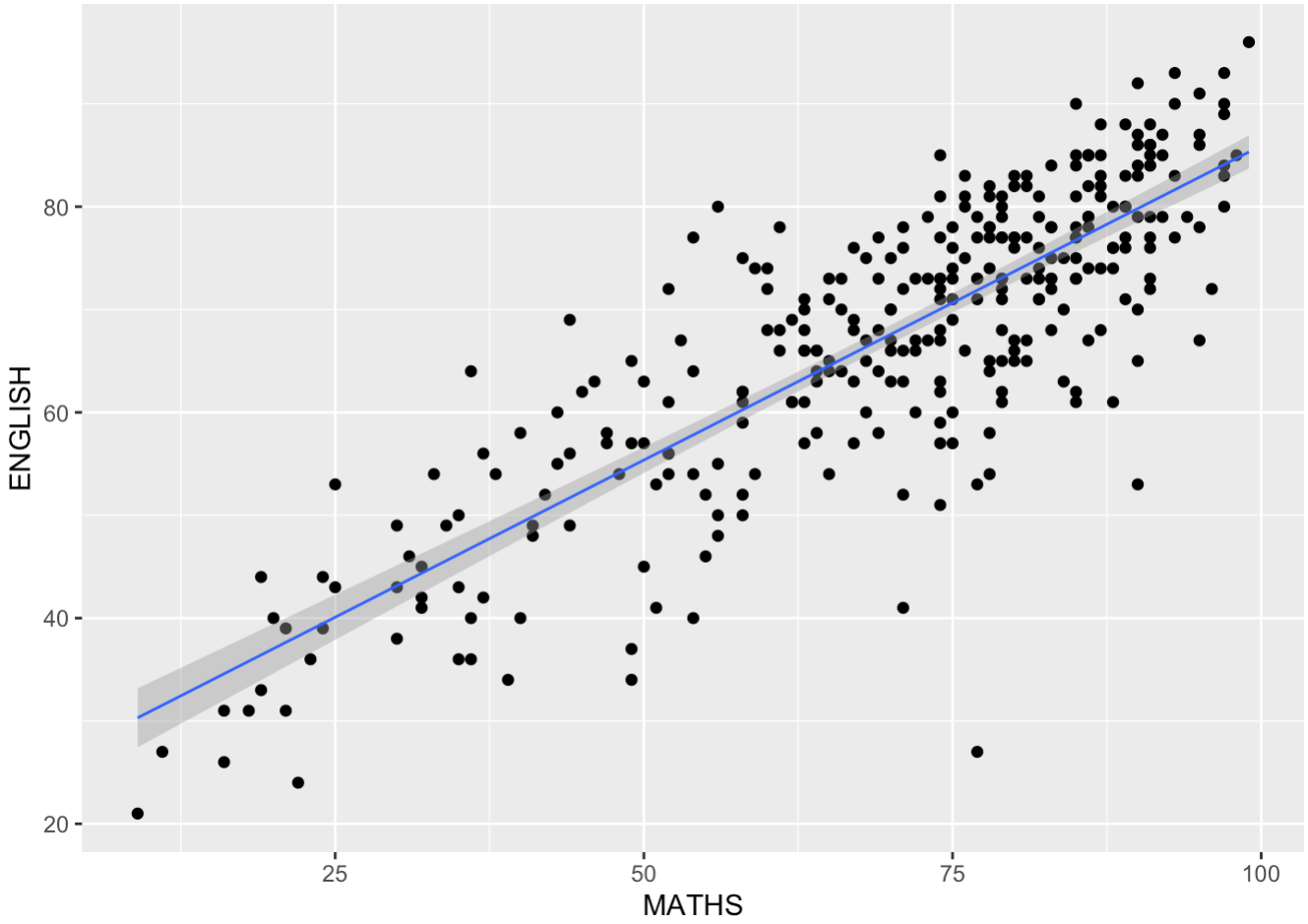
Flipped bar chart (horizontal)

```
ggplot(data=exam_data,
       aes(x=RACE)) +
  geom_bar() +
  coord_flip()
```



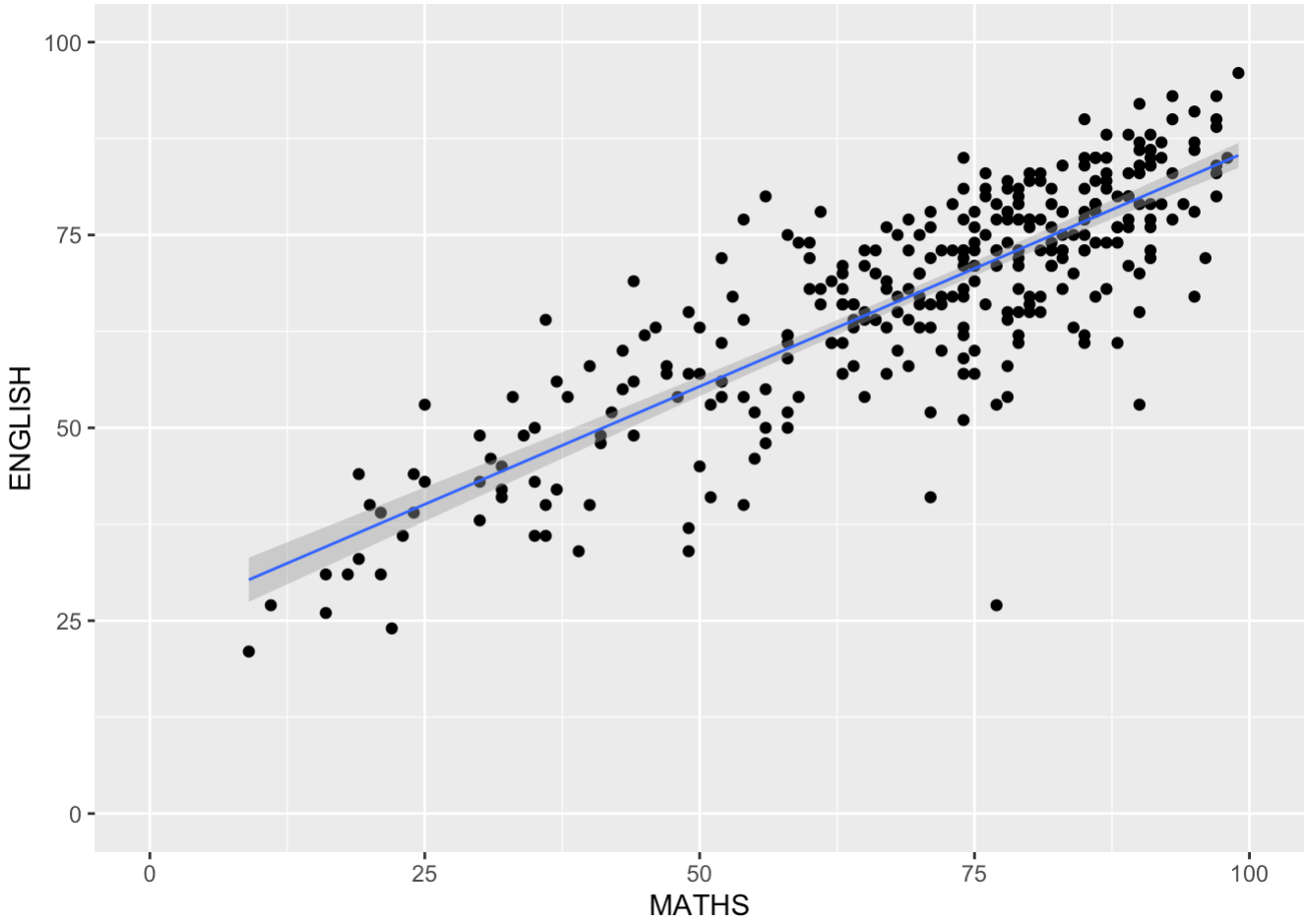

```
ggplot(data=exam_data,
       aes(x= MATHS, y=ENGLISH)) +
  geom_point() +
  geom_smooth(method=lm, size=0.5)
```

```
`geom_smooth()` using formula = 'y ~ x'
```

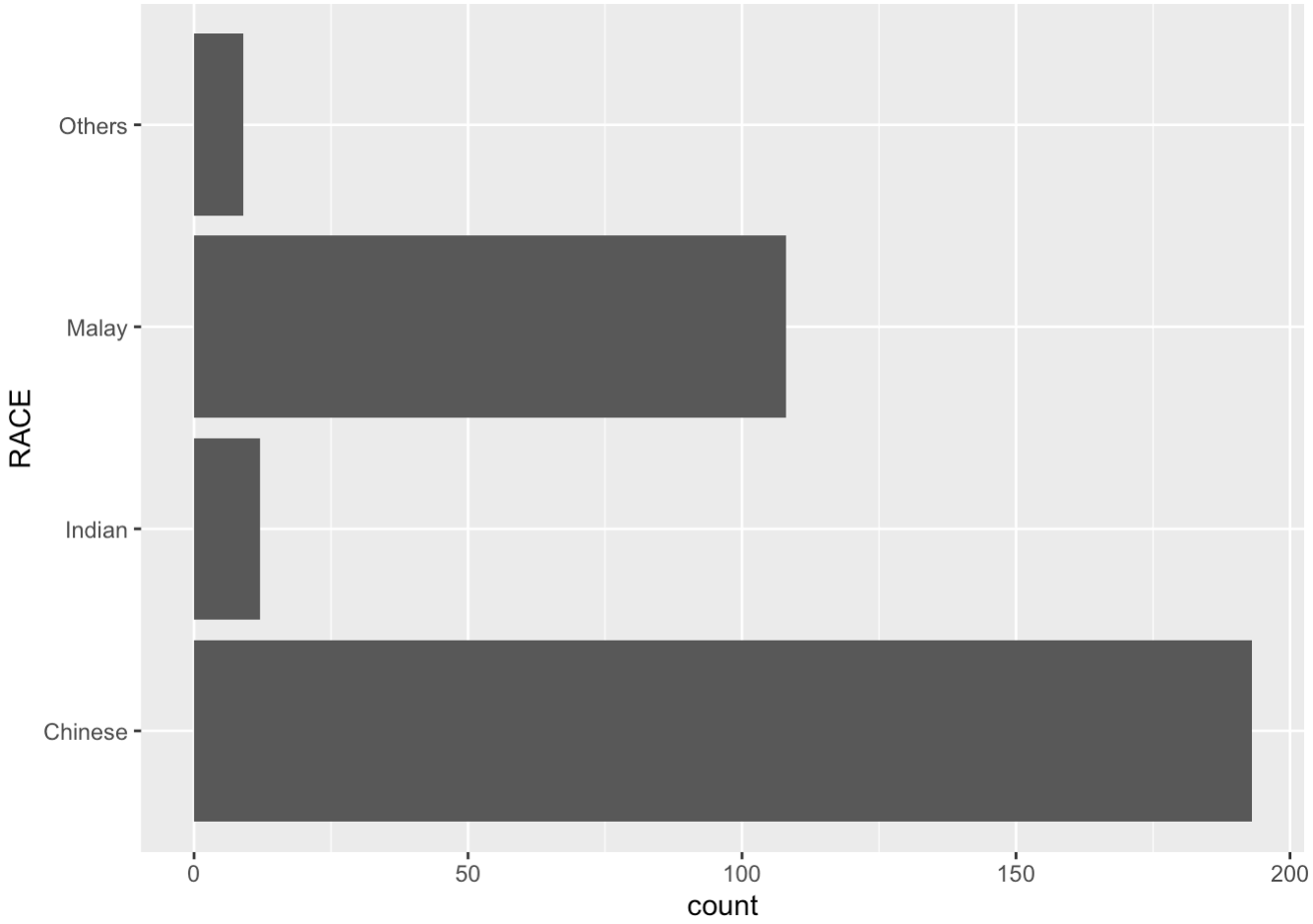



```
ggplot(data=exam_data,
       aes(x= MATHS, y=ENGLISH)) +
  geom_point() +
  geom_smooth(method=lm,
              size=0.5) +
  coord_cartesian(xlim=c(0,100),
                  ylim=c(0,100))
```

```
`geom_smooth()` using formula = 'y ~ x'
```

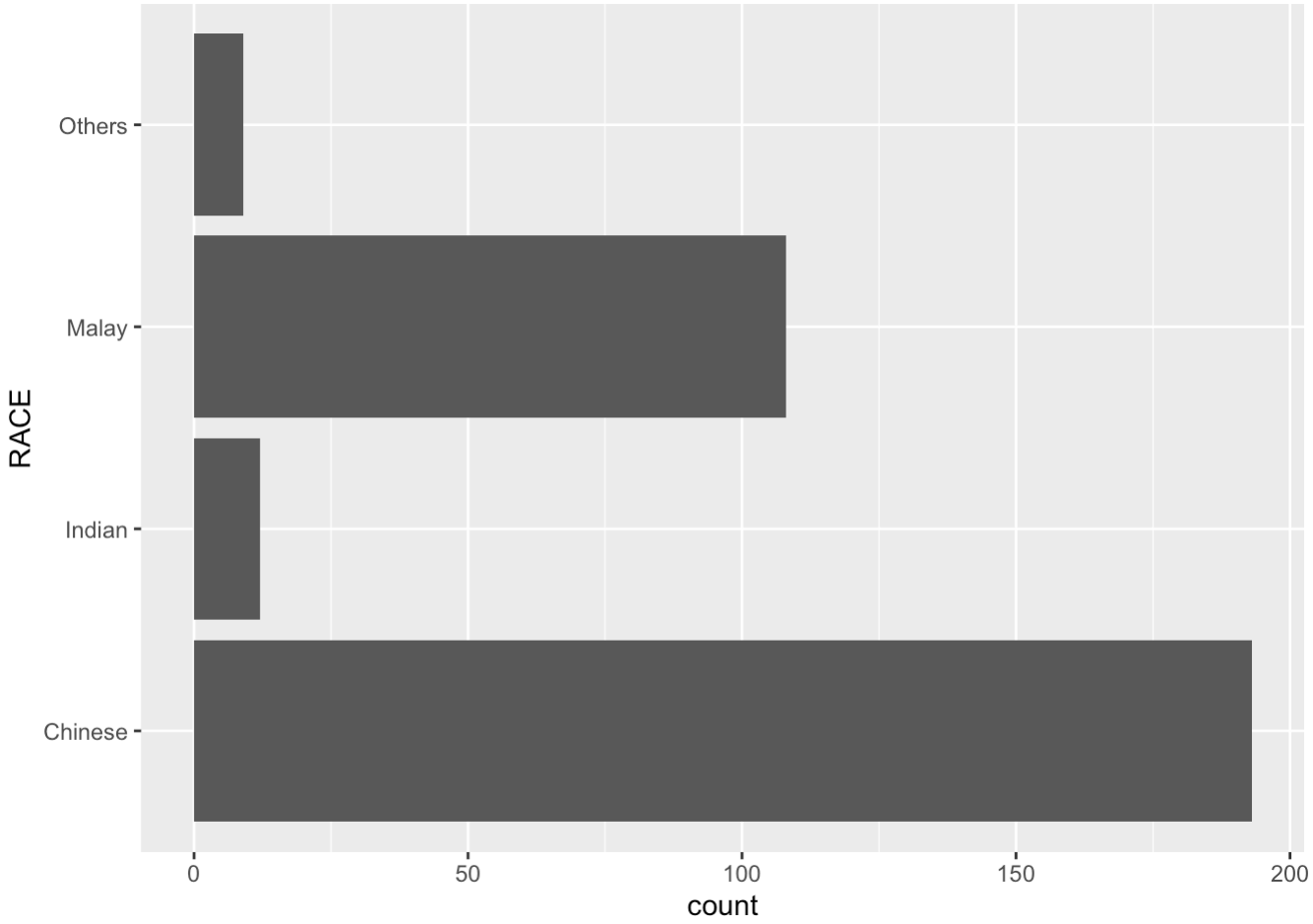



```
ggplot(data=exam_data,
       aes(x=RACE)) +
  geom_bar() +
  coord_flip() +
  theme_gray()
```



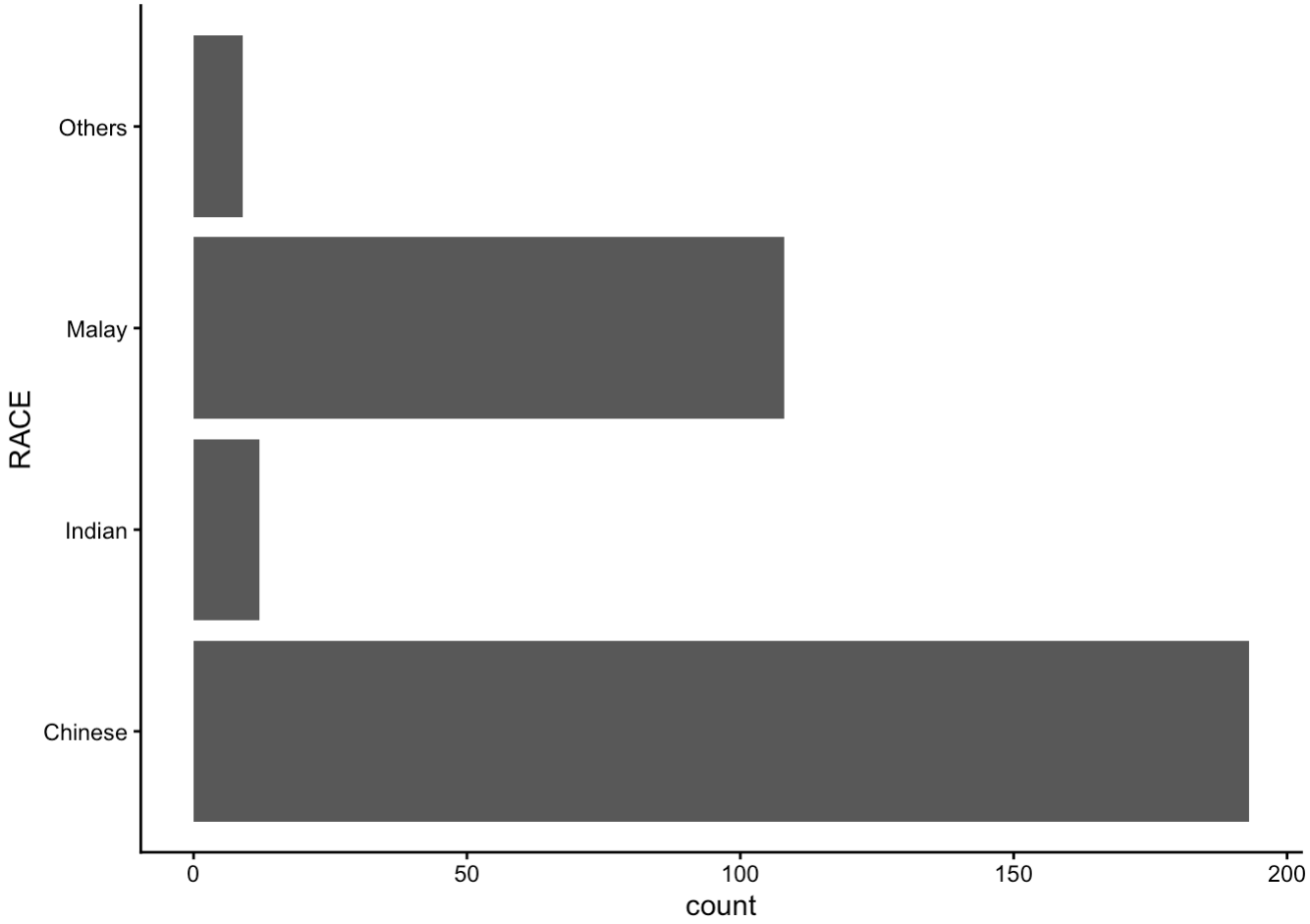
Bar Chart using theme_gray

```
ggplot(data=exam_data,
       aes(x=RACE)) +
  geom_bar() +
  coord_flip() +
  theme_gray()
```



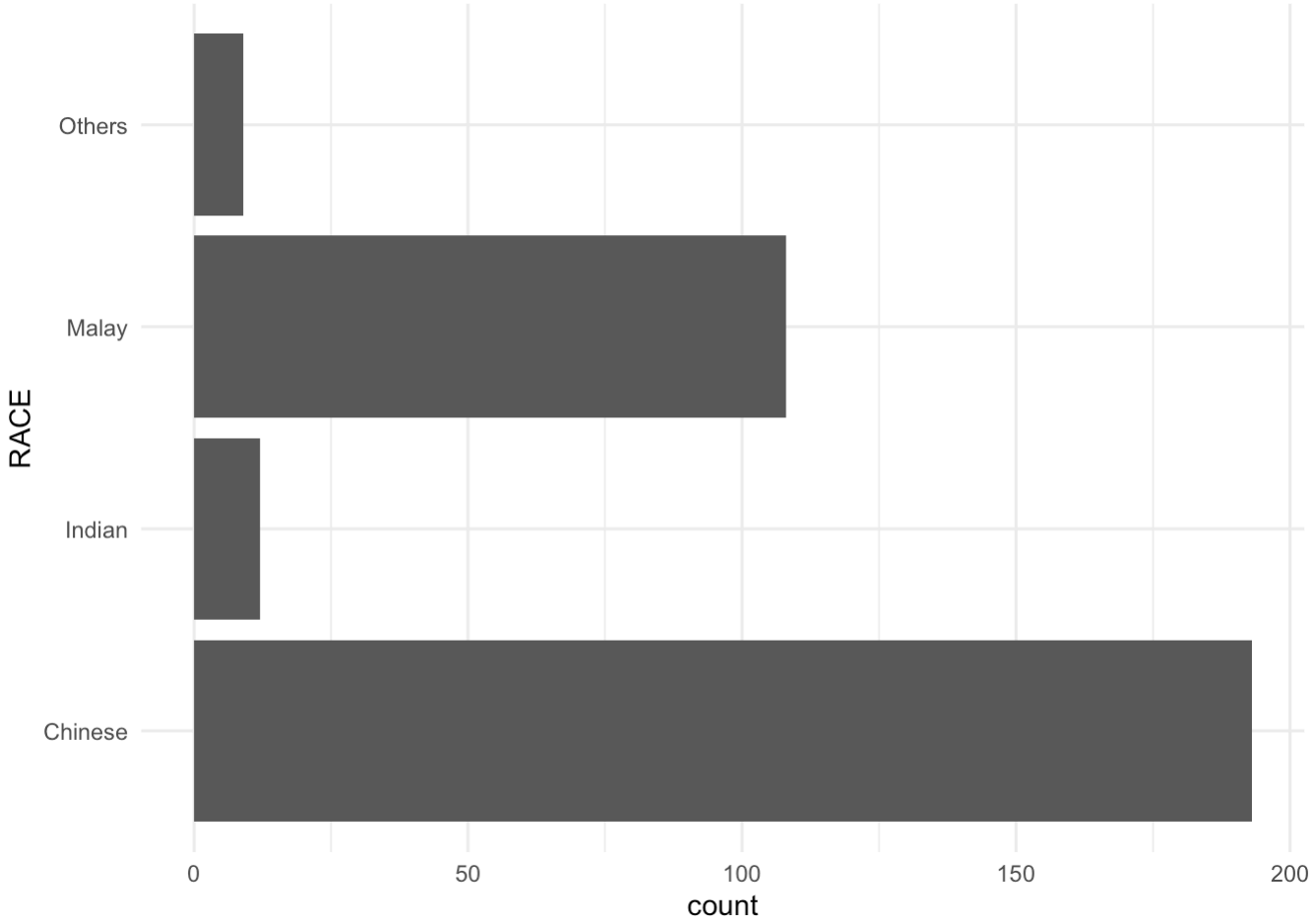
Bar Chart using theme_classic

```
ggplot(data=exam_data,
       aes(x=RACE)) +
  geom_bar() +
  coord_flip() +
  theme_classic()
```



Bar Chart using theme minimal

```
ggplot(data=exam_data,
       aes(x=RACE)) +
  geom_bar() +
  coord_flip() +
  theme_minimal()
```


Others
Malay
Indian
Chinese
RACE
0
50
100
150
200
count